

Drug Summary v2 · Recall Sheet

Every drug class from pharmacology 1 and 2, in the order they are taught — class, the drugs under it, and a short indication. **Cover the right strip of one column and name the drugs from memory.** The full reference (mechanism, ADRs, interactions) is the v1 document.

	Exam-critical	the fact that actually gets tested - the one to know cold	895×
	Dangerous	life-threatening, or the toxicity that defines the drug	581×
	Caution	the nuance that trips people up - a contraindication or a trap	413×
	Explicitly safe	stated as safe where you would expect it not to be (e.g. pregnancy)	14×

Contents · สารบัญ

PHARMACOLOGY 1

4

CYP450 INDUCERS & INHIBITORS

4

Enzyme Inducers — "GPRS Cell Phone" 4
Enzyme Inhibitors — "SICK FACES.COM" 4

ADRENERGIC DRUGS

4

Non-Selective Direct Agonists 4
α1-Selective Agonists 4
α2-Selective Agonists — Centrally-Acting Sympatholytics 5
β1-Selective Agonists 5
β2-Selective Agonists 5
β3-Selective Agonists 5
Dopamine-Receptor Agonists 5
Indirect & Mixed-Acting Sympathomimetics 5

Selective α1-Blockers 5
Non-Selective α-Blockers 5
Non-Selective β-Blockers 5
Cardioselective (β1) Blockers 5
β-Blockers with ISA 5
α + β Blockers (Third-Generation) 5

CHOLINERGIC AGENTS

6

Direct-Acting Cholinergic Agonists 6
Reversible Cholinesterase Inhibitors 6
Irreversible Cholinesterase Inhibitors (Organophosphates) 6
Cholinesterase-Inhibitor Poisoning — Treatment 6
Ganglionic Blockers 6
Muscarinic Antagonists (Antimuscarinics) 6
Neuromuscular Blockers — Non-Depolarizing 6

DRUGS AFFECTING GI MOTILITY

6

Bulk-Forming Laxatives 6
Osmotic Laxatives 6
Stool-Wetting Laxatives & Emollients 7
Chloride Secretion Activators 7
Stimulant Laxatives 7
IBAT Inhibitors 7
Oral Rehydration Therapy 7
Absorbents, Bulk-Formers & Mucosal Coaters 7
Enkephalinase Inhibitors 7
Opioid Anti-Motility Agents 7
Prokinetics — D2 Antagonists 7
Prokinetics — 5-HT4 Agonists 7
Prokinetics — Cholinergic Enhancers 7
Anti-Spasmotics 7

DRUGS FOR ACID-PEPTIC DISEASES

7

Antacids 7
Anti-flatulences, Digestive Enzymes & Anti-reflux Agents 7
H₂ Antagonists 8
Proton Pump Inhibitors (PPIs) 8
Potassium-Competitive Acid Blockers (PCABs) 8
Misoprostol 8
Mucus-Enhancing Agents 8
Sucralfate 8
Bismuth-Containing Drugs 8

DRUGS FOR INFLAMMATORY BOWEL DISEASE

8

5-Aminosalicylates (5-ASA) 8
Corticosteroids 8
Thiopurines 8
Methotrexate 8
Anti-TNF-α Biologics 8
Anti-Integrin (Vedolizumab) 8
Anti-IL-12/23 (Ustekinumab) 8
JAK Inhibitor (Tofacitinib) 8

PROSTANOIDS, NSAIDS & LEUKOTRIENE ANTAGONISTS

8

Non-selective NSAIDs (tNSAIDs) 9
COX-2 Selective Inhibitors (Coxibs) 9
Prostanoid Analogues 9
Leukotriene Modifiers 9
PAF Antagonists 9

DRUGS FOR GOUT

9

Colchicine 9
NSAIDs 9
Corticosteroids 9
IL-1 Inhibitors 9
Xanthine Oxidase Inhibitors 9
Uricosurics 9
Uricase 10

DRUGS IN OA & RA

10

Acetaminophen 10
NSAIDs (oral) 10
Topical Agents 10
Tramadol 10
Duloxetine 10
Intra-articular Agents 10
Glucosamine, Chondroitin & DMOADs 10
Glucocorticoids (RA) 10
csDMARDs — Methotrexate 10
csDMARDs — Sulfasalazine · Leflunomide · Hydroxychloroquine 10
bDMARDs — TNF Inhibitors 10
bDMARDs — Interleukin Inhibitors 10
bDMARDs — B-cell & T-cell Targeted 10
tsDMARDs — JAK Inhibitors 10

HISTAMINE, ANTIHISTAMINES & SEROTONIN AGONISTS

11

1st-Generation H1 Antihistamines 11
2nd-Generation H1 Antihistamines 11
Betahistine (H3) 11
Triptans — 5-HT_{1B/1D} Agonists 11
Ergots 11

DIURETICS

11

Carbonic Anhydrase Inhibitor 11
Thiazide & Thiazide-like 11
Loop Diuretic 11
K⁺-Sparing Diuretic 11
Osmotic Diuretic 11
Vasodilators 12
Hydralazine 12
Minoxidil — K⁺ Channel Opener 12
Sodium Nitroprusside 12
Organic Nitrates 12
PDE-5 Inhibitors 12
Endothelin Receptor Antagonists (ERAs) 12
sGC Stimulators 12
Neprilysin Inhibitor / ARNI 12

CALCIUM CHANNEL BLOCKERS

12

Dihydropyridines (DHP) 12
Phenylalkylamine — Verapamil 12
Benzothiazepine — Diltiazem 12
Other Non-Dihydropyridines 12

THE RENIN-ANGIOTENSIN SYSTEM

12

ACE Inhibitor 13
Angiotensin Receptor Blocker (ARB) 13
Direct Renin Inhibitor (DRI) 13
ARNI — Angiotensin Receptor–Neprilysin Inhibitor 13

ANTIHYPERTENSIVE AGENTS

13

Diuretics in Hypertension 13
Centrally Acting Sympatholytics 13
Adrenergic Neuron Blockers 13
α- and Mixed αβ-Blockers 13
β-Blockers in Hypertension 13
Direct Vasodilators 13

ANTIANGINAL AGENTS

13

Organic Nitrates 14
Calcium Channel Blockers (CCBs) 14
β-Blockers 14
Ivabradine — Direct Bradycardic Agent 14
Ranolazine — Late Na⁺ Current Inhibitor 14
Trimetazidine — Metabolic Modulator 14

ANTIARRHYTHMIC DRUGS

14

Class IA — Na⁺ Block, Intermediate Dissociation 14
Class IB — Na⁺ Block, Fast Dissociation 14
Class IC — Na⁺ Block, Slow Dissociation 14
Class II — β-Adrenergic Antagonists 14
Class III — K⁺ Channel Blockers 14
Class IV — Ca²⁺ Channel Blockers 14

PHARMACOLOGY OF HEART FAILURE

14

Cardiac Glycosides 14
β₁-Adrenergic Agonists 15
Phosphodiesterase Inhibitors 15
Calcium Sensitizer 15
Loop Diuretics 15
K⁺-Sparing Diuretics & MRA 15
Nitrovasodilators (NO Donors) 15
Hydralazine + Nitrate 15
ACE Inhibitors 15
Angiotensin Receptor Blockers 15
ARNI (Angiotensin Receptor–Neprilysin Inhibitor) 15
Nesiritide (Recombinant BNP) 15
SGLT2 Inhibitors 15
β-Adrenergic Blockers in HF 15
Ivabradine 15
Other & Emerging Agents 15

ANTITHROMBOTICS

15

COX-1 Inhibitors (Aspirin) 16
P2Y₁₂ Antagonists 16
GPIIb/IIIa Antagonists 16
PDE Inhibitors 16
PAR-1 Antagonists 16
Heparins & Fondaparinux 16
Warfarin 16
DOACs / NOACs 16
Thrombolytics / Fibrinolytics 16

LIPID-LOWERING AGENTS

16

Statins (HMG-CoA Reductase Inhibitors) 16
Bempedoic Acid (ATP–Citrate Lyase Inhibitor) 16
PCSK9 Inhibitors 16
Ezetimibe (Cholesterol Absorption Inhibitor) 16
Bile Acid Sequestrants 16
Fibrates (Fibric Acid Derivatives) 17
Niacin (Nicotinic Acid) 17
Omega-3 Fatty Acids 17

PHARMACOLOGY 2

17

PRINCIPLES OF ANTIBIOTIC USE

17

Empiric, Definitive & Prophylactic Therapy 17
Combination Therapy 17

DRUGS AFFECTING CELL WALL SYNTHESIS

17

Natural Penicillins 17
Penicillinase-Resistant Penicillins 17
Aminopenicillins 17
Antipseudomonal Penicillins 17
β-Lactamase Inhibitors & Combinations 17
Cephalosporins — 1st Generation 18
Cephalosporins — 2nd Generation 18
Cephalosporins — 3rd Generation 18
Cephalosporins — 4th & 5th Generation 18
Carbapenems 18

GLYCOPEPTIDES & LIPOGLYCOPEPTIDES

18

Glycopeptides 18
Lipoglycopeptides 18
Bacitracin 18

DRUGS AFFECTING THE CELL MEMBRANE

18

Polymyxins 18
Daptomycin 18

INHIBITORS OF DNA, NUCLEIC ACID & PROTEIN SYNTHESIS

18

Fluoroquinolones 19
Metronidazole 19
Macrolides 19
Ketolides (Telithromycin) 19
Lincosamides (Lincomycin & Clindamycin) 19
Streptogramins (Quinupristin / Dalfopristin) 19

Chloramphenicol 19
Aminoglycosides 19
Tetracyclines 19
Glycylcyclines (Tigecycline) 19
Oxazolidinones (Linezolid) 19
Folate Synthesis Inhibitors (TMP-SMX) 19

ANTIMYCOBACTERIAL DRUGS

19

Isoniazid (INH) 19
Rifamycins 19
Pyrazinamide 19
Ethambutol 19
Aminoglycosides *(second-line)* 20
Fluoroquinolones *(second-line)* 20
Oral Bacteriostatic Second-Line Agents 20
Bedaquiline 20
Nitroimidazoles — Pretomanid & Delamanid 20
Linezolid 20
Dapsone 20
Clofazimine 20

ANTIFUNGAL DRUGS

20

Polyenes 20
Allylamines 20
Azoles — Imidazoles 20
Azoles — Triazoles 20
Echinocandins 20
Triterpenoids 20
Flucytosine 20
Griseofulvin 21

ANTIPARASITIC DRUGS

21

Aminoquinolines — Chloroquine, Amodiaquine, Piperaquine 21
Quinine & Quinidine 21
Mefloquine 21
Artemisinin & Derivatives (ACTs) 21
Other Heme-Polymerisation Inhibitors 21
8-Aminoquinolines — Primaquine & Tafenoquine 21
Atovaquone 21
Antimalarial Translation Inhibitors 21
Antimalarial Folate Inhibitors 21
Nitroimidazoles — Metronidazole & Tinidazole 21
Pentamidine 21
Anti-Toxoplasma Agents 21
Benzimidazoles — Albendazole & Mebendazole 21
Ivermectin 21
Neuromuscular Paralyzing Anthelmintics 21
Praziquantel 21
Niclosamide 21
Diethylcarbamazine (DEC) 22

ANTI-HEPATITIS AGENTS

22

Interferons (IFN-α / PegIFN) 22
HBV Nucleoside / Nucleotide Analogues (NAs) 22
Ribavirin 22
HCV NS5B Polymerase Inhibitors 22
HCV NS5A Inhibitors 22
HCV NS3/4A Protease Inhibitors 22

ANTIVIRALS & ANTI-HIV DRUGS

22

Antitherpes Agents 22
Anti-CMV Agents 22
Influenza — M2 Proton Channel Inhibitors 22
Influenza — Neuraminidase Inhibitors 22
Influenza — Polymerase Complex Inhibitors 22
Other Antivirals 22
NRTIs — Nucleoside/Nucleotide RT Inhibitors 23
NNRTIs — Non-Nucleoside RT Inhibitors 23
Protease Inhibitors (PIs, -navir) 23
Entry Inhibitors 23
INSTIs — Integrase Strand Transfer Inhibitors (-gravir) 23

ANTI-DIABETIC DRUGS

23

Insulin 23
GLP-1 Receptor Agonists 23
Dual GIP/GLP-1 Receptor Agonist 23
DPP-4 Inhibitors 23
Sulfonylureas 24
Meglitinides (Non-sulfonylureas) 24
Biguanide — Metformin 24
Thiazolidinediones (TZDs) 24
α-Glucosidase Inhibitors 24
SGLT2 Inhibitors 24

ESTROGENS, PROGESTINS & CONTRACEPTIVES	24	ANTICANCER AGENTS II — TARGETED THERAPY	30
Estrogens	24	EGFR Inhibitors — Small-Molecule TKIs	30
Selective Estrogen Receptor Modulators (SERMs)	24	EGFR Inhibitors — Monoclonal Antibodies	30
Estrogen Receptor Antagonists	24	HER2 Inhibitors	30
Aromatase Inhibitors	24	BCR-ABL Inhibitors	30
Progestins	24	mTOR Inhibitors	30
Antiprogestins	24	BRAF Inhibitors	30
Progestin-Only Contraceptives (POC)	25	PARP Inhibitors	30
Emergency Contraception	25	Angiogenesis (VEGF) Inhibitors	30
THYROID HORMONES & ANTITHYROID DRUGS	25	Cancer Vaccines	30
Thyroid Hormone Replacement	25	Immune Checkpoint Inhibitors	30
Thionamides	25	Anti-Estrogens — SERMs & SERDs	31
Anion Inhibitors	25	Aromatase Inhibitors	31
Iodides	25	Anti-Androgens	31
Radioactive Iodine (¹³¹ I)	25	Gene Expression Modulators	31
Adjunctive & Thyroid Storm Therapy	25	IMMUNOMODULATORS	31
DRUGS REGULATING BONE-MINERAL HOMEOSTASIS	25	Induction Antibodies	31
Vitamin D Derivatives	26	B-Cell Targeted Therapy (Desensitization)	31
Bisphosphonates	26	Corticosteroids	31
RANKL Inhibitors	26	Calcineurin Inhibitors	31
Estrogens, STEARs & SERMs	26	Antimetabolites	31
Calcitonin	26	mTOR Inhibitors	31
Parathyroid Hormone Analogues	26	Costimulation Blocker	31
Sclerostin Inhibitors	26	JAK Inhibitors	31
Vitamin K Analogues & Fluoride	26	Specific Immunoglobulins	31
Calcimimetics	26	Myeloid Growth Factors & Erythropoietins	31
Phosphate Binders	26	Interferons & Imiquimod	31
HYPOTHALAMIC-PITUITARY HORMONES	26	Immunomodulatory Imides	32
GnRH Agonists	26	CAR T-Cell Therapy	32
GnRH Antagonists	26	PHARMACOLOGY 3	32
Gonadotropins	27	DRUGS ACTING ON GABA-A RECEPTOR	32
Dopamine Agonists	27	Benzodiazepines	32
Oxytocin & Oxytocin Antagonists	27	Barbiturates	32
Growth Hormone	27	Z-Drugs	32
Somatostatin Analogs	27	Flumazenil	32
Vasopressin & Analogs	27	ANALGESICS I — OPIOIDS & PARACETAMOL	32
Vasopressin Antagonists	27	Weak Opioids	32
ANABOLIC-ANDROGENIC STEROIDS & ANTI-ANDROGENS	27	Strong Opioids	32
Testosterone & Testosterone Esters	27	Partial Agonists & Mixed Agonist–Antagonists	32
17 α -Alkylated Testosterone	27	Opioid Antagonists	32
Anabolic Steroids	27	Paracetamol Overdose & Antidote	32
Steroid Synthesis Inhibitors	27	CNS STIMULANTS AND DRUGS AFFECTING APPETITE	33
5 α -Reductase Inhibitors	27	DRUG ABUSE AND ADDICTION	33
Steroidal Androgen Receptor Antagonists	28	ANXIOLYTICS AND HYPNOTICS	33
Non-Steroidal Androgen Receptor Antagonists	28	Antidepressants for Anxiety	33
ADRENAL STEROIDS & THEIR ANTAGONISTS	28	Benzodiazepines in Anxiety	33
Mineralocorticoids	28	Pregabalin & Gabapentin	33
Mineralocorticoid Receptor Antagonists	28	Buspirone	33
Glucocorticoids — Systemic	28	Other Agents for Anxiety	33
Glucocorticoids — Topical & Local	28	Melatonin & Melatonin Receptor Agonists	33
Glucocorticoid Antagonists	28	Orexin Receptor Antagonists (DORAs)	33
ANTICANCER AGENTS I	28	ANTIDEPRESSANTS AND LITHIUM	33
Alkylating Agents	29	MAO Inhibitors (MAOIs and RIMAs)	34
Platinum Analogs	29	Tricyclic Antidepressants (TCAs)	34
Antifolates	29	SSRIs	34
Ribonucleotide Reductase Inhibitor	29	SARIs — Serotonin Antagonist and Reuptake Inhibitors	34
Pyrimidine Analogs	29	SNRIs	34
Purine Analogs	29	NDRI	34
Vinca Alkaloids	29	NaSSAs	34
Taxanes	29	Lithium	34
Topoisomerase I Inhibitors	29	ANTIPSYCHOTICS	34
Topoisomerase II Inhibitors	29	Conventional (First-Generation) Antipsychotics	35
Anthracyclines	29	Extrapyramidal Side Effects	35
Bleomycin & Mitomycin C	29	Atypical (Second-Generation) Antipsychotics	35
L-Asparaginase	29		

LOCAL AND GENERAL ANESTHETICS	35
NEURODEGENERATIVE DISEASE DRUGS	35
Cholinesterase Inhibitors (ChEIs)	35
NMDA Receptor Antagonist — Memantine	35
Anti-Amyloid Monoclonal Antibodies	35
Levodopa + Dopa Decarboxylase Inhibitor	35
MAO-B Inhibitors	35
COMT Inhibitors	35
Dopamine Receptor Agonists	36
Amantadine & Anticholinergics	36
ANTIEPILEPTICS	36
Na ⁺ Channel Blockers	36
Ca ²⁺ Channel Blockers	36
GABA Enhancers	36
Broad Spectrum — Valproate	36
Broad Spectrum — Topiramate	36
AMPA Receptor Blocker — Perampanel	36
SV2A Modulators — Levetiracetam	36
MIGRAINE HEADACHE	36
Ergots	37
Triptans & Lasmiditan	37
Gepants — CGRP Receptor Antagonists	37
Anti-CGRP Monoclonal Antibodies	37
ANALGESICS II — TWO SEPARATE TERRITORIES	37
Sodium Channel Blockers	37
Gabapentinoids ($\alpha_2\delta$ -1 Ligands)	37
Antidepressants — TCAs & SNRIs	37
Opioids — Tramadol & Methadone	37
Other Agents — Capsaicin, Ketamine, Cannabinoids	37
Orphenadrine	37
Tolperisone & Eperisone	37
Benzodiazepines — Diazepam & Clonazepam	37
Baclofen	37
Tizanidine	37
Riluzole	38

Pharmacology 1

ภกป 304 · ภาคต้น · autonomic → GI → inflammation → cardiovascular

CYP450 Inducers & Inhibitors

2 classes · 20 drugs

Enzyme Inducers — "GPRS Cell Phone"

(7)

Griseofulvin	dermatophyte
Phenytoin	epilepsy
Rifampicin	tuberculosis
St. John's wort	herbal — used for depression
Carbamazepine	epilepsy
Phenobarbitone	epilepsy
Cigarette smoke ★	not a drug — counts...

Enzyme Inhibitors — "SICK FACES.COM"

(13)

Sulphonamides	bacterial infection
Antifungals — itraconazole · ketoconazole · fluconazole	fungal infection
Macrolides — clarithromycin · azithromycin · erythromycin	bacterial infection
Ciprofloxacin	bacterial infection
Sertraline	depression
Cimetidine	peptic ulcer / GERD
Omeprazole	peptic ulcer / GERD
Metronidazole	anaerobic bacteria
Antivirals — ritonavir · indinavir · nelfinavir · amprenavir	HIV
Antiarrhythmics — amiodarone · quinidine	arrhythmia
Antidepressants — fluoxetine · paroxetine	depression
Isoniazid	tuberculosis
Alcohol	

Adrenergic Drugs

14 classes · 54 drugs

Non-Selective Direct Agonists

(4)

Epinephrine (α_1 α_2 β_1 β_2) ★	anaphylactic shock — drug of...
Norepinephrine (α_1 α_2 β_1, little β_2)	vasopressor — cardiogenic shock, hypotension
Isoproterenol (β_1 = β_2)	inotropic agent
Oxymetazoline · Xylometazoline · Tetrahydrozoline (non-selective α)	topical nasal decongestant

α_1 -Selective Agonists

(2)

Phenylephrine	nasal decongestant
Midodrine (prodrug → desglymidodrine)	orthostatic hypotension — the orally...

α2-Selective Agonists — Centrally-Acting Sympatholytics (7)

Clonidine	hypertension
Methyldopa (→ α-methyl-NE in brain) ★	hypertension — the choice in...
Guanfacine	hypertension
Guanabenz	hypertension
Moxonidine (α2 + imidazoline I1)	hypertension
Tizanidine	muscle relaxant
Apraclonidine · Brimonidine	glaucoma — topical, no BP...

β1-Selective Agonists (1)

Dobutamine	acute heart failure
-------------------	---------------------

β2-Selective Agonists (5)

Salbutamol	asthma, obstructive airway disease
Terbutaline	asthma, COPD, status asthmaticus
Salmeterol / Formoterol	asthma maintenance; formoterol also exercise-induced...
Indacaterol / Vilanterol	COPD
Ritodrine	uterine relaxant — largely abandoned,...

β3-Selective Agonists (1)

Mirabegron	overactive bladder
-------------------	--------------------

Dopamine-Receptor Agonists (2)

Dopamine	cardiogenic shock with low cardiac...
Fenoldopam (D1 >> D2)	severe hypertension in hospital —...

Indirect & Mixed-Acting Sympathomimetics (6)

Amphetamine · Methamphetamine · Dextroamphetamine (displacers)	CNS stimulant, appetite suppression —...
Methylphenidate (displacer)	ADHD
Cocaine (NET inhibitor)	local anaesthetic with vasoconstriction; otherwise...
Atomoxetine (NET inhibitor)	ADHD
Ephedrine (mixed)	decongestant
Pseudoephedrine (mixed)	OTC oral decongestant

Selective α1-Blockers (7)

Prazosin	hypertension
Terazosin	hypertension
Doxazosin	hypertension
Alfuzosin	BPH
Tamsulosin (α1A)	BPH — least postural hypotension
Silodosin (α1A)	BPH
Indoramin · Urapidil	hypertension

Non-Selective α-Blockers (3)

Phenoxybenzamine (α1 > α2, irreversible / covalent, slow onset, long-lasting)	pheochromocytoma
Phentolamine (α1 = α2, competitive, rapid and transient)	hypertensive emergency in pheochromocytoma
Yohimbine · Tolazoline (α2 >> α1)	little clinical use — orthostatic...

Non-Selective β-Blockers (4)

Propranolol ★	hypertension
Nadolol	hypertension
Timolol	open-angle glaucoma
Pindolol · Carteolol · Penbutolol	hypertension

Cardioselective (β1) Blockers (7)

Atenolol	hypertension
Metoprolol	hypertension
Bisoprolol	hypertension
Acebutolol	hypertension
Esmolol	IV only — perioperative tachycardia,...
Betaxolol	hypertension
Celiprolol · Nebivolol	hypertension — also vasodilating

β-Blockers with ISA (3)

Pindolol	hypertension — the classic ISA...
Acebutolol	hypertension
Carteolol · Celiprolol	hypertension

α + β Blockers (Third-Generation) (2)

Labetalol ★	hypertensive emergency
Carvedilol ★	heart failure with reduced ejection...

Cholinergic Agents

7 classes · 35 drugs

Direct-Acting Cholinergic Agonists (6)

Acetylcholine	topical only — induction of...
Bethanechol ★	urinary retention
Carbachol	glaucoma
Pilocarpine ★	glaucoma
Cevimeline	dry mouth / dry eyes...
Methacholine	diagnostic only

Reversible Cholinesterase Inhibitors (5)

Edrophonium ★	diagnosis of myasthenia gravis —...
Neostigmine	myasthenia gravis
Pyridostigmine	myasthenia gravis — longer-acting than...
Physostigmine	glaucoma / ↓ intraocular pressure...
Donepezil · Rivastigmine · Galantamine	mild–moderate Alzheimer's disease

Irreversible Cholinesterase Inhibitors (Organophosphates) (3)

Parathion · Malathion · Ecothiopate	insecticides — the real-world poisoning...
Dyflos (isofluorophate)	experimental / ophthalmic
Sarin	nerve gas

Cholinesterase-Inhibitor Poisoning — Treatment (4)

Decontamination	activated charcoal
Supportive	airway, ventilation — respiratory failure...
Antimuscarinic ★	Atropine IV
Reactivators	Pralidoxime

Ganglionic Blockers (3)

Hexamethonium · Trimethaphan · Mecamylamine	classic Nn ganglion blockers
Botulinum toxin	blocks nerve-stimulated ACh release —...
Hemicholinium	blocks choline transport into the...

Muscarinic Antagonists (Antimuscarinics) (8)

Atropine ★	sinus bradycardia
Hyoscine (scopolamine)	motion sickness
Tropicamide · Cyclopentolate	mydriasis for eye exam
Ipratropium · Tiotropium ★	asthma & COPD
Benzhexol (trihexyphenidyl) · Benztropine	parkinsonism — the CNS-penetrating ones
Dicycloverine (dicyclomine)	antispasmodic in irritable bowel syndrome
Oxybutynin · Tolterodine · Trospium	overactive bladder — non-selective
Solifenacin · Darifenacin	overactive bladder / urinary incontinence...

Neuromuscular Blockers — Non-Depolarizing (6)

Tubocurarine	surgical muscle relaxation — slow
Pancuronium	surgical muscle relaxation, long cases...
Vecuronium	surgical muscle relaxation
Atracurium ★	surgical muscle relaxation in hepatic...
Mivacurium	muscle relaxation for short procedures...
Rocuronium ★	rapid-sequence intubation — the alternative...

Drugs Affecting GI Motility

14 classes · 36 drugs

Bulk-Forming Laxatives (1)

Psyllium (ispaghula husk)	constipation
----------------------------------	--------------

Osmotic Laxatives (4)

Magnesium hydroxide	constipation
Lactulose	constipation
Polyethylene glycol (PEG / macrogol) ★	low dose = constipation
Sodium phosphate	occasional constipation

Stool-Wetting Laxatives & Emollients (3)

Glycerin	constipation — local, rectal
Docusate (dioctyl sodium sulfosuccinate)	constipation — often combined with...
Mineral oil (liquid paraffin)	constipation

Chloride Secretion Activators (1)

Lubiprostone	chronic constipation
---------------------	----------------------

Stimulant Laxatives (3)

Bisacodyl	constipation
Senna	constipation
Castor oil	constipation

IBAT Inhibitors (1)

Elobixibat	chronic constipation
-------------------	----------------------

Oral Rehydration Therapy (1)

Oral rehydration salts (ORS)	prevention & treatment of dehydration...
-------------------------------------	--

Absorbents, Bulk-Formers & Mucosal Coaters (5)

Activated charcoal	diarrhoea
Kaolin (kaolinite clay)	diarrhoea — usually with pectin
Pectin	diarrhoea — reduces stool fluidity
Cholestyramine (bile-salt binding resin)	bile-salt diarrhoea
Dioctahedral smectite	diarrhoea — absorbent plus mucosal...

Enkephalinase Inhibitors (1)

Racecadotril	acute diarrhoea — secretory component
---------------------	---------------------------------------

Opioid Anti-Motility Agents (2)

Loperamide	acute & chronic non-infectious diarrhoea
Diphenoxylate	diarrhoea — not currently available...

Prokinetics — D2 Antagonists (3)

Domperidone	dyspepsia
Metoclopramide	dyspepsia
Itopride	dyspepsia

Prokinetics — 5-HT₄ Agonists (3)

Mosapride	dyspepsia
Prucalopride	constipation — a pure 5-HT ₄ ...
Cisapride	withdrawn in many countries —...

Prokinetics — Cholinergic Enhancers (1)

Acotiamide	dyspepsia
-------------------	-----------

Anti-Spasmodics (7)

Hyoscine-N-butylbromide (anti-muscarinic)	abdominal colic
Clidinium bromide (anti-muscarinic, + chlordiazepoxide)	IBS pain
Dicyclomine (anti-muscarinic, + simethicone)	IBS pain, with bloating
Mebeverine (smooth-muscle relaxant)	IBS pain
Alverine (+ simethicone)	IBS pain, with bloating
Drotaverine	visceral spasm
Pinaverium bromide	IBS pain

Drugs for Acid-Peptic Diseases

9 classes · 24 drugs

Antacids (4)

NaHCO₃	rapid symptomatic relief
Mg(OH)₂	rapid relief — laxative pull
CaCO₃	slower, longer relief
Al(OH)₃	slower, longer relief — constipating

Anti-flatulences, Digestive Enzymes & Anti-reflux Agents (4)

Simethicone	bloating
Pancreatin (± Aspergillus oryzae enzymes)	pancreatic insufficiency
Alginate / sodium alginate	GERD — used combined with...
Activated charcoal	claimed for gas — efficacy...

H₂ Antagonists (2)

Cimetidine	peptic ulcers
Famotidine	peptic ulcers

Proton Pump Inhibitors (PPIs) (6)

Omeprazole	ulcers
Esomeprazole	ulcers
Lansoprazole	ulcers
Dexlansoprazole	GERD
Pantoprazole	ulcers
Rabeprazole ★	ulcers

Potassium-Competitive Acid Blockers (PCABs) (1)

Vonoprazan	acid-peptic disease where speed/depth of...
-------------------	---

Misoprostol (1)

Misoprostol	prevention of NSAID-induced peptic ulcers
--------------------	---

Mucus-Enhancing Agents (3)

Rebamipide (TID)	gastric ulcers
Teprenone (TID)	gastric ulcers
Irsogladine (OD)	gastric ulcers

Sucralfate (1)

Sucralfate	peptic ulcers
-------------------	---------------

Bismuth-Containing Drugs (2)

Bismuth subsalicylate	peptic ulcers — especially in...
Bismuth subcitrate / subnitrate / subcarbonate	peptic ulcers

Drugs for Inflammatory Bowel Disease

8 classes · 19 drugs

5-Aminosalicylates (5-ASA) (4)

Sulfasalazine	mild–moderate UC — 5-ASA +...
Mesalamine (5-ASA)	mild–moderate UC — the active...
Olsalazine	UC — 2 mesalamine molecules...
Balsalazide	UC — mesalamine + 4-ABA...

Corticosteroids (6)

Prednisolone (oral)	active UC/CD not responding to...
Hydrocortisone (IV)	severe disease
Methylprednisolone (IV)	severe disease
Budesonide MMX	mild–moderate left-sided/extensive UC
Ileal-release budesonide	mild–moderate ileocecal CD
Beclomethasone dipropionate	active left-sided/extensive UC

Thiopurines (2)

Azathioprine	maintenance of remission in UC...
Mercaptopurine (6-MP)	maintenance of remission in UC...

Methotrexate (1)

Methotrexate ★	maintenance of remission in CD...
-----------------------	-----------------------------------

Anti-TNF-α Biologics (3)

Infliximab	moderate–severe UC and CD
Adalimumab	moderate–severe UC and CD
Golimumab	moderate–severe UC

Anti-Integrin (Vedolizumab) (1)

Vedolizumab	moderate–severe UC
--------------------	--------------------

Anti-IL-12/23 (Ustekinumab) (1)

Ustekinumab	moderate–severe CD — induction and...
--------------------	---------------------------------------

JAK Inhibitor (Tofacitinib) (1)

Tofacitinib	moderate–severe UC, oral — induction...
--------------------	---

Prostanoids, NSAIDs & Leukotriene Antagonists

5 classes · 22 drugs

Non-selective NSAIDs (tNSAIDs) (7)

Ibuprofen	mild–moderate pain
Naproxen	inflammatory pain, OA/RA — the...
Diclofenac	inflammatory pain, migraine COX-2- leaning →...
Indomethacin	acute gout
Mefenamic acid	dysmenorrhoea, mild–moderate pain
Piroxicam / meloxicam	chronic arthritis — long $t_{1/2}$,...
Ketorolac	short-term moderate–severe post- operative pain

COX-2 Selective Inhibitors (Coxibs) (3)

Celecoxib	OA/RA and inflammatory pain where...
Etoricoxib	OA/RA, acute gout, inflammatory pain
Parecoxib	injectable prodrug of valdecoxib —...

Prostanoid Analogues (7)

Alprostadil (PGE₁)	erectile dysfunction — relaxes corpus...
Lubiprostone (PGE₁)	constipation — activates a Cl [−] ...
Misoprostol (PGE₁) ★	mucoprotection
Dinoprostone (PGE₂)	oxytocic — cervical ripening /...
Latanoprost, bimatoprost, travoprost (PGF_{2α})	glaucoma — ↑ outflow of...
Carboprost (PGF_{2α})	oxytocic
Epoprostenol, iloprost, treprostinil (PGI₂)	pulmonary arterial hypertension — relaxes...

Leukotriene Modifiers (4)

Montelukast (CysLTR1 antagonist, $t_{1/2}$ 2.7–7 h)	asthma
Pranlukast (CysLTR1 antagonist, $t_{1/2}$ 3–9 h)	asthma
Zafirlukast (CysLTR1 antagonist, $t_{1/2}$ 10 h)	asthma
Zileuton (5-LOX inhibitor, $t_{1/2}$ 2.5 h)	asthma

PAF Antagonists (1)

Rupatadine	allergic rhinitis
-------------------	-------------------

Drugs for Gout

7 classes · 20 drugs

Colchicine (1)

Colchicine	acute gout flare
-------------------	------------------

NSAIDs (3)

Naproxen	acute flare
Indomethacin	acute flare
Diclofenac	acute flare

Corticosteroids (4)

Prednisolone	acute flare — 30–35 mg/day...
Betamethasone	acute flare — single IM...
Methylprednisolone	acute flare — taper over...
Triamcinolone acetonide	acute flare — single IM...

IL-1 Inhibitors (3)

Anakinra	frequent flares when the first-line...
Rilonacept	same
Canakinumab	same

Xanthine Oxidase Inhibitors (2)

Allopurinol	first-line ULT in chronic gout...
Febuxostat	second-line — allopurinol failure, intolerance,...

Uricosurics (5)

Probenecid	chronic gout — inhibits URAT1,...
Benzbromarone	chronic gout — URAT1 + ...
Sulfinpyrazone	withdrawn in many countries — ...
Lesinurad	URAT1 + OAT4; approved 2015,...
Dotinurad	newest agent

Uricase (2)

Pegloticase	refractory / severe tophaceous gout...
Rasburicase	withdrawn for gout

Drugs in OA & RA

14 classes · 35 drugs

Acetaminophen (1)

Acetaminophen (paracetamol)	first-line mild–moderate OA pain
------------------------------------	----------------------------------

NSAIDs (oral) (6)

Ibuprofen (non-selective)	moderate–severe OA pain
Naproxen (non-selective) ★	same — lower CV risk,...
Diclofenac (non-selective)	moderate–severe OA pain
Meloxicam (non-selective)	RA symptoms
Celecoxib (COX-2 selective)	OA / RA when GI...
Etoricoxib (COX-2 selective)	OA / RA when GI...

Topical Agents (3)

Diclofenac gel (1–3%) ★	first-line mild–moderate knee and hand...
Ketoprofen gel (2.5%)	same
Topical capsaicin	localized pain — elbow, knee,...

Tramadol (1)

Tramadol	moderate–severe OA pain — second-line...
-----------------	--

Duloxetine (1)

Duloxetine	chronic musculoskeletal pain — knee...
-------------------	--

Intra-articular Agents (2)

IA corticosteroid (triamcinolone, methylprednisolone)	significant local inflammation & pain...
IA hyaluronic acid	knee OA — second-line, when...

Glucosamine, Chondroitin & DMOADs (3)

Glucosamine	mild–moderate OA — 500 mg...
Chondroitin	mild–moderate OA — 200–400 mg...
DMOADs	structural modification in OA —...

Glucocorticoids (RA) (2)

Prednisolone (oral)	acute RA inflammation — 5–10...
Triamcinolone / methylprednisolone (intra-articular)	a local flare in one...

csDMARDs — Methotrexate (1)

Methotrexate (MTX) ★	first-line DMARD for RA —...
-----------------------------	------------------------------

csDMARDs — Sulfasalazine · Leflunomide · Hydroxychloroquine (3)

Sulfasalazine ★	RA — second choice; preferred...
Leflunomide	RA — efficacy similar to...
Hydroxychloroquine	RA — cheap, moderate efficacy;...

bDMARDs — TNF Inhibitors (5)

Etanercept	RA after csDMARD failure
Adalimumab	RA after csDMARD failure
Infliximab	RA after csDMARD failure
Golimumab	RA after csDMARD failure
Certolizumab	RA after csDMARD failure

bDMARDs — Interleukin Inhibitors (2)

Tocilizumab (IL-6 receptor)	RA after csDMARD failure
Anakinra (IL-1 receptor antagonist)	RA after csDMARD failure —...

bDMARDs — B-cell & T-cell Targeted (2)

Rituximab (anti-CD20)	RA — 1000 mg IV...
Abatacept (T-cell co-stimulation blocker)	RA — SC weekly or...

tsDMARDs — JAK Inhibitors (3)

Tofacitinib	RA — oral targeted therapy
Baricitinib	RA — oral targeted therapy
Upadacitinib	RA — oral targeted therapy

Histamine, Antihistamines & Serotonin Agonists

5 classes · 25 drugs

1st-Generation H1 Antihistamines

(7)

Chlorpheniramine (CPM)	allergic rhinitis
Diphenhydramine	allergy
Dimenhydrinate	motion sickness, vertigo
Promethazine	allergy
Hydroxyzine	urticaria & itchy skin disorders...
Cyproheptadine	appetite stimulation — via its...
Doxepin ★	chronic urticaria — ~800× more...

2nd-Generation H1 Antihistamines

(8)

Cetirizine	allergic rhinitis
Levocetirizine	as cetirizine — the active...
Loratadine	allergic rhinitis
Desloratadine	as loratadine — the active...
Fexofenadine	allergic rhinitis
Bilastine	allergic rhinitis — no hepatic...
Rupatadine · Ebastine · Mizolastine · Acrivastine	allergic rhinitis
Azelastine · Olopatadine · Levocabastine · Emedastine · Epinastine	topical — conjunctival / nasal...

Betahistine (H3)

(1)

Betahistine	vertigo / vestibular disorders
--------------------	--------------------------------

Triptans — 5-HT_{1B/1D} Agonists

(8)

Sumatriptan	acute migraine — t _{1/2} 2...
Rizatriptan	acute migraine — t _{1/2} 2...
Zolmitriptan	acute migraine — t _{1/2} 3...
Almotriptan	acute migraine — t _{1/2} 3.5...
Eletriptan	acute migraine — t _{1/2} 5...
Naratriptan	acute migraine — t _{1/2} 5–6.3...
Frovatriptan ★	acute migraine — t _{1/2} 25...
Lasmiditan (ditan)	acute migraine — selective 5-HT _{1F} ...

Ergots

(1)

Ergotamine	acute migraine
-------------------	----------------

Diuretics

5 classes · 15 drugs

Carbonic Anhydrase Inhibitor

(3)

Acetazolamide	glaucoma
Brinzolamide	glaucoma
Dorzolamide	glaucoma

Thiazide & Thiazide-like

(3)

Hydrochlorothiazide (HCTZ)	hypertension
Indapamide	hypertension
Chlorthalidone	hypertension

Loop Diuretic

(4)

Furosemide	edema
Torsemide	edema
Bumetanide	edema
Ethacrynic acid	edema — the non-sulfonamide option

K⁺-Sparing Diuretic

(4)

Spironolactone (ARA)	hypertension
Eplerenone (ARA)	heart failure
Amiloride (NCC)	hypertension
Triamterene (NCC)	edema

Osmotic Diuretic

(1)

Mannitol	prevent oliguria / acute renal...
-----------------	-----------------------------------

Vasodilators

8 classes · 21 drugs

Hydralazine (1)

Hydralazine	hypertensive emergency in pregnancy
--------------------	-------------------------------------

Minoxidil — K⁺ Channel Opener (3)

Minoxidil	severe hypertension refractory to other...
Pinacidil	
Nicorandil	

Sodium Nitroprusside (1)

Sodium nitroprusside	hypertensive emergencies
-----------------------------	--------------------------

Organic Nitrates (4)

Nitroglycerin (NTG)	angina pectoris
Isosorbide dinitrate (ISDN)	angina pectoris
Isosorbide 5-mononitrate (ISMN)	angina pectoris
Pentaerythritol tetranitrate (PETN)	angina pectoris

PDE-5 Inhibitors (4)

Sildenafil	erectile dysfunction
Tadalafil ★	ED
Vardenafil	ED
Avanafil	ED

Endothelin Receptor Antagonists (ERAs) (3)

Ambrisentan — selective ETA, 200:1	PAH
Bosentan — dual ETA/ETB, 20:1	PAH
Macitentan — dual ETA/ETB, 50:1	PAH

sGC Stimulators (2)

Riociguat	pulmonary arterial hypertension
Vericiguat	chronic symptomatic heart failure with...

Neprilysin Inhibitor / ARNI (3)

Sacubitril / valsartan (ARNI)	heart failure, NYHA class II–IV...
Sacubitril (alone)	not effective as monotherapy
Candoxatril	

Calcium Channel Blockers

4 classes · 14 drugs

Dihydropyridines (DHP) (9)

Nifedipine	hypertension
Amlodipine	hypertension
Felodipine	hypertension
Nicardipine	hypertension
Lercanidipine	hypertension — nephroprotective, less peripheral...
Manidipine (L/T-type)	hypertension — less vasodilatory edema
Cilnidipine (L/N-type)	hypertension — also blocks sympathetic...
Lacidipine	hypertension
Nimodipine	ischemic neurological deficit after subarachnoid...

Phenylalkylamine — Verapamil (1)

Verapamil	paroxysmal SVT
------------------	----------------

Benzothiazepine — Diltiazem (1)

Diltiazem	angina
------------------	--------

Other Non-Dihydropyridines (3)

Bepidil (diarylaminopropylamine)	refractory angina
Cinnarizine (diphenylpiperazine)	class listed only — no...
Flunarizine (diphenylpiperazine)	class listed only — no...

The Renin–Angiotensin System

4 classes · 14 drugs

ACE Inhibitor (6)

Captopril	hypertension
Enalapril	hypertension
Lisinopril	hypertension
Ramipril · Benazepril · Quinapril	hypertension
Perindopril · Trandolapril · Moexipril	hypertension — long-acting prodrugs
Fosinopril	hypertension — the phosphorus-containing one

Angiotensin Receptor Blocker (ARB) (6)

Losartan	hypertension
Valsartan	hypertension
Candesartan	hypertension
Irbesartan	hypertension
Telmisartan	hypertension — longest t _{1/2}
Azilsartan · Olmesartan · Eprosartan	hypertension

Direct Renin Inhibitor (DRI) (1)

Aliskiren	hypertension — a low-molecular-weight nonpeptide;...
------------------	--

ARNI — Angiotensin Receptor–Neprilysin Inhibitor (1)

Valsartan/sacubitril (LCZ696)	heart failure with reduced ejection...
--------------------------------------	--

Antihypertensive Agents

6 classes · 23 drugs

Diuretics in Hypertension (3)

Hydrochlorothiazide (HCTZ)	mild–moderate HT — 10–15 mmHg...
Furosemide	severe HT
Spironolactone	added to avoid excessive K ⁺ ...

Centrally Acting Sympatholytics (4)

Methyldopa ★	hypertension in pregnancy — the...
Clonidine	resistant HT not responding to...
Guanabenz	hypertension
Guanfacine	hypertension

Adrenergic Neuron Blockers (2)

Guanethidine	hypertension — peripheral only
Reserpine	hypertension — an alkaloid from...

α- and Mixed αβ-Blockers (5)

Prazosin	hypertension — α ₁ selective
Doxazosin	hypertension
Terazosin	hypertension
Labetalol	hypertension — mixed, β ₁ =...
Carvedilol	hypertension — mixed

β-Blockers in Hypertension (3)

Propranolol · nadolol · timolol	HT — non-selective
Metoprolol · atenolol · alprenolol ★	HT — β ₁ -selective, possibly safer...
Pindolol · acebutolol · penbutolol	HT — with ISA; ↓BP...

Direct Vasodilators (6)

Hydralazine	hypertension
Minoxidil	hypertension
Nitroprusside	hypertension — acts on vascular...
Diazoxide	hypertension — acts on vascular...
Fenoldopam	hypertension — peripheral arteriolar dilation
Trimethaphan	hypertension — the odd one...

Antianginal Agents

6 classes · 18 drugs

Organic Nitrates (4)

Nitroglycerin (NTG)	acute attack
Isosorbide dinitrate (ISDN)	acute attack
Isosorbide 5-mononitrate (5-ISMN)	prophylaxis — the oral nitrate
Pentaerythritol tetranitrate (PETN)	prophylaxis, oral

Calcium Channel Blockers (CCBs) (6)

Amlodipine (DHP)	angina
Nifedipine (DHP)	angina
Nicardipine (DHP)	angina
Felodipine (DHP)	hypertension
Diltiazem (non-DHP)	angina
Verapamil (non-DHP)	angina

β-Blockers (5)

Propranolol, timolol, nadolol, alprenolol (non-selective)	angina
Pindolol, carteolol, penbutolol, acebutolol (ISA)	not for severe angina —...
Labetalol, carvedilol (non-selective + α₁)	added vasodilation
Atenolol, metoprolol, bisoprolol, esmolol (β₁-selective)	stable & unstable angina
Nebivolol (β₁-selective + NO)	added vasodilation

Ivabradine — Direct Bradycardic Agent (1)

Ivabradine	anginal attack — efficacy similar...
-------------------	--------------------------------------

Ranolazine — Late Na⁺ Current Inhibitor (1)

Ranolazine	angina — US FDA approved,...
-------------------	------------------------------

Trimetazidine — Metabolic Modulator (1)

Trimetazidine	stable angina — demonstrated efficacy
----------------------	---------------------------------------

Antiarrhythmic Drugs

6 classes · 20 drugs

Class IA — Na⁺ Block, Intermediate Dissociation (3)

Quinidine	conversion of atrial flutter/fibrillation +...
Procainamide	PVCs
Disopyramide	PVCs

Class IB — Na⁺ Block, Fast Dissociation (4)

Lidocaine	drug of choice for acute...
Mexiletine	ventricular arrhythmia
Tocainide	ventricular arrhythmia
Phenytoin	ventricular arrhythmia

Class IC — Na⁺ Block, Slow Dissociation (3)

Flecainide	sustained ventricular tachycardia
Propafenone	sustained ventricular tachycardia
Encainide	ventricular arrhythmia

Class II — β-Adrenergic Antagonists (3)

Propranolol ★	supraventricular tachyarrhythmias — reduces sudden...
Acebutolol	premature ventricular beats
Esmolol	acute arrhythmias in surgery /...

Class III — K⁺ Channel Blockers (5)

Amiodarone	recurrent ventricular fibrillation
Dronedarone	atrial fibrillation
Sotalol	ventricular & supraventricular arrhythmias
Ibutilide	atrial fibrillation / flutter conversion
Dofetilide	atrial fibrillation / flutter

Class IV — Ca²⁺ Channel Blockers (2)

Verapamil	supraventricular tachycardia
Diltiazem	supraventricular tachycardia

Pharmacology of Heart Failure

16 classes · 35 drugs

Cardiac Glycosides (2)

Digoxin	symptomatic HFrEF
----------------	-------------------

Digitoxin	as above
------------------	----------

β₁-Adrenergic Agonists (3)

Dobutamine	acute decompensated HF — continuous...
Dopamine	inotropic support
Epinephrine / Norepinephrine	inotropic + pressor support

Phosphodiesterase Inhibitors (2)

Milrinone	acute decompensated HF — constant...
Amrinone	acute decompensated HF

Calcium Sensitizer (1)

Levosimendan	decompensated heart failure
---------------------	-----------------------------

Loop Diuretics (4)

Furosemide	HF congestion
Bumetanide	edema
Torsemide	edema
Ethacrynic acid	edema

K⁺-Sparing Diuretics & MRA (3)

Spironolactone ★	the only diuretic that improves...
Amiloride	edema — no survival benefit
Triamterene	edema — no survival benefit

Nitrovasodilators (NO Donors) (3)

Nitroglycerin	acute HF / congestion —...
Isosorbide dinitrate	oral nitrate — t _{1/2} 1–4...
Sodium nitroprusside	acute HF — continuous IV...

Hydralazine + Nitrate (1)

Hydralazine (+ isosorbide dinitrate)	HF in patients who cannot...
---	------------------------------

ACE Inhibitors (2)

Enalapril · Lisinopril · Ramipril	heart failure
Captopril · Benazepril · Quinapril · Fosinopril · Trandolapril	heart failure

Angiotensin Receptor Blockers (3)

Losartan	heart failure
Valsartan	heart failure
Candesartan	heart failure

ARNI (Angiotensin Receptor–Neprilysin Inhibitor) (1)

Sacubitril/valsartan (LCZ696)	HFrEF — the ARNI pillar...
--------------------------------------	----------------------------

Nesiritide (Recombinant BNP) (1)

Nesiritide	acute decompensated HF — continuous...
-------------------	--

SGLT2 Inhibitors (2)

Dapagliflozin	HFrEF
Empagliflozin	HFrEF

β-Adrenergic Blockers in HF (4)

Metoprolol succinate	HFrEF — β ₁ -selective
Bisoprolol	HFrEF — β ₁ -selective
Nebivolol	HFrEF — β ₁ -selective with NO-potentiating...
Carvedilol	HFrEF — β ₁ , β ₂ and...

Ivabradine (1)

Ivabradine	stable chronic HFrEF, LVEF ≤35%,...
-------------------	-------------------------------------

Other & Emerging Agents (2)

Vericiguat — soluble guanylate cyclase stimulator	LVEF <45%, NYHA II–IV; second-line...
Omecamtiv mecarbil — selective cardiac myosin activator ●	development halted by the manufacturer

Antithrombotics

9 classes · 25 drugs

COX-1 Inhibitors (Aspirin) (2)

Aspirin (ASA), low-dose	arterial thrombosis — primary &...
Triflusal	same space; COX-1 inhibition is...

P2Y12 Antagonists (5)

Ticlopidine	the first thienopyridine — displaced...
Clopidogrel	PCI
Prasugrel	ACS-PCI
Ticagrelor	ACS
Cangrelor (IV only)	PCI

GPIIb/IIIa Antagonists (3)

Abciximab	Fab fragment of a humanised...
Eptifibatide	cyclic heptapeptide
Tirofiban	tyrosine derivative — competitive; 60–70%...

PDE Inhibitors (2)

Dipyridamole (PDE-3 and PDE-5)	secondary prevention of stroke
Cilostazol (potent PDE-3)	secondary prevention of stroke

PAR-1 Antagonists (1)

Vorapaxar	secondary prevention after prior MI...
------------------	--

Heparins & Fondaparinux (4)

UFH	acute anticoagulation where titration and...
Enoxaparin	VTE
Nadroparin, Dalteparin	VTE prophylaxis & treatment
Fondaparinux	VTE — the option when...

Warfarin (1)

Warfarin	VTE
-----------------	-----

DOACs / NOACs (3)

Dabigatran etexilate	the only oral direct thrombin...
Desirudin · Lepirudin · Bivalirudin · Argatroban	injectable direct thrombin inhibitors
Rivaroxaban · Apixaban · Edoxaban · Betrixaban	direct FXa inhibitors — VTE,...

Thrombolytics / Fibrinolytics (4)

Streptokinase	ACS
Alteplase (recombinant human tPA)	acute ischaemic stroke
Tenecteplase	acute ischaemic stroke
Reteplase ●	ACS — contraindicated in acute...

Lipid-lowering Agents

8 classes · 22 drugs

Statins (HMG-CoA Reductase Inhibitors) (7)

Rosuvastatin	hypercholesterolemia — high-intensity, ↓LDL 63%
Atorvastatin	hypercholesterolemia — high-intensity, ↓LDL 50%
Pitavastatin ★	hypercholesterolemia — ↓LDL 48%
Simvastatin	hypercholesterolemia — ↓LDL 41%
Pravastatin	hypercholesterolemia — ↓LDL 34%
Lovastatin	hypercholesterolemia — ↓LDL 34%
Fluvastatin	hypercholesterolemia — ↓LDL 24%

Bempedoic Acid (ATP-Citrate Lyase Inhibitor) (1)

Bempedoic acid ★	hypercholesterolemia — especially the statin-intolerant...
-------------------------	--

PCSK9 Inhibitors (3)

Evolocumab (mAb)	↑ LDL-C — ↓LDL 64%
Alirocumab (mAb)	↑ LDL-C — ↓LDL 58%
Inclisiran (siRNA) ★	↑ LDL-C — dosed day...

Ezetimibe (Cholesterol Absorption Inhibitor) (1)

Ezetimibe	↑ LDL-C — usually added...
------------------	----------------------------

Bile Acid Sequestrants (3)

Cholestyramine	↑ LDL-C
Colestipol	↑ LDL-C
Colesevelam	↑ LDL-C

Fibrates (Fibric Acid Derivatives) (4)

Fenofibrate ★	hypertriglyceridemia — better than gemfibrozil...
Gemfibrozil ●	hypertriglyceridemia — t½ 1–2 h
Bezafibrate	hypertriglyceridemia — t½ 1–2 h
Clofibrate	hypertriglyceridemia

Niacin (Nicotinic Acid) (1)

Niacin (crystalline)	hypertriglyceridemia
-----------------------------	----------------------

Omega-3 Fatty Acids (2)

Icosapent ethyl	hypertriglyceridemia + residual CV risk...
EPA/DHA ethyl esters (fish oil)	hypertriglyceridemia — 3–4 g/d with...

Pharmacology 2

ภกปว 305 • ภาคปลาย • anti-infectives → endocrine → oncology → immune

Principles of Antibiotic Use

2 classes • 6 drugs

Empiric, Definitive & Prophylactic Therapy (3)

Gastroenteritis	ciprofloxacin or norfloxacin or tetracycline
Wound infection	
Septicaemia	

Combination Therapy (3)

Trimethoprim + sulfamethoxazole	blocks two steps in the...
β-lactam + β-lactamase inhibitor	inhibits enzymatic inactivation of the...
Penicillin/ampicillin + gentamicin/streptomycin	cell-wall damage increases aminoglycoside uptake

Drugs Affecting Cell Wall Synthesis

10 classes • 40 drugs

Natural Penicillins (2)

Penicillin G	Streptococcus spp.
Penicillin V	Streptococcus spp.

Penicillinase-Resistant Penicillins (6)

Methicillin	Staphylococcus aureus
Nafcillin	same
Oxacillin	same
Cloxacillin	same
Dicloxacillin	same
Flucloxacillin	same

Aminopenicillins (4)

Ampicillin	Listeria monocytogenes
Amoxicillin ★	same — the one penicillin...
Bacampicillin	same
Ciclacillin	same

Antipseudomonal Penicillins (6)

Piperacillin	aminopenicillin spectrum + P. aeruginosa
Ticarcillin	same
Carbenicillin	same
Azlocillin	same
Mezlocillin	same
Temocillin	same

β-Lactamase Inhibitors & Combinations (6)

Amoxicillin/clavulanate	streptococci
Ampicillin/sulbactam	same, + A. baumannii
Piperacillin/tazobactam	same, + P. aeruginosa + ...
Cefoperazone/sulbactam	Enterobacteriaceae
Ceftolozane/tazobactam	Enterobacteriaceae incl. some ESBL
Ceftazidime/avibactam	Enterobacteriaceae incl. some ESBL and...

Cephalosporins — 1st Generation (3)

Cefazolin	streptococci
Cephalexin	streptococci
Cefadroxil · Cephalexin · Cephalexin · Cephalexin	same generation; the lecture lists...

Cephalosporins — 2nd Generation (2)

Cefuroxime	streptococci
Cefaclor · Cefamandole · Cefoxitin · Cefmetazole · Cefotetan	same generation; the lecture lists...

Cephalosporins — 3rd Generation (5)

Ceftriaxone ★	streptococci
Cefotaxime	same — also reaches the...
Cefixime · Cefdinir · Cefditoren	the oral 3rd-generation options
Ceftazidime	CA-Enterobacteriaceae
Cefoperazone/sulbactam	Enterobacteriaceae

Cephalosporins — 4th & 5th Generation (2)

Ceftaroline ★	gram-positive cocci including MRSA
Cefepime · Cefpirome · Ceftobiprole	listed in the class chart;...

Carbapenems (4)

Imipenem/cilastatin	Enterobacteriaceae incl. ESBL
Meropenem	same range, + imipenem-resistant P...
Ertapenem	Enterobacteriaceae including most ESBL
Doripenem · Biapenem · Panipenem	other members of the class

Glycopeptides & Lipoglycopeptides

3 classes · 7 drugs

Glycopeptides (3)

Vancomycin (IV)	MSSA
Vancomycin (oral) ★	C. difficile only — it...
Teicoplanin (IV or IM)	same spectrum as vancomycin

Lipoglycopeptides (3)

Telavancin (from vancomycin)	complicated skin & soft-tissue infection
Dalbavancin (from teicoplanin) ★	complicated skin & soft-tissue infection
Oritavancin	named by the lecture as...

Bacitracin (1)

Bacitracin (topical)	Gram-positive skin infection — S...
-----------------------------	-------------------------------------

Drugs Affecting the Cell Membrane

2 classes · 3 drugs

Polymyxins (2)

Polymyxin B	Gram-negative infection — multidrug-resistant P...
Colistin (polymyxin E)	same — MDR Gram-negative infection

Daptomycin (1)

Daptomycin	serious Gram-positive infection — MRSA/VRSA
-------------------	---

Inhibitors of DNA, Nucleic Acid & Protein Synthesis

12 classes · 31 drugs

Fluoroquinolones (5)

Nalidixic acid (1st)	the parent quinolone — all...
Norfloxacin · Ofloxacin · Ciprofloxacin · Rufloxacin (2nd)	gram-negative infections
Levofloxacin (3rd)	respiratory + urinary infections, now...
Moxifloxacin · Sitafoxacin · Prulifloxacin (4th)	respiratory & skin infections
Gemifloxacin · Gatifloxacin	S. pneumoniae respiratory infections

Metronidazole (1)

Metronidazole ★	anaerobic infections incl. brain abscess
------------------------	--

Macrolides (5)

Erythromycin	the natural parent — corynebacterial...
Clarithromycin	H. pylori triple therapy
Azithromycin	CAP & AECB
Roxithromycin	URI
Spiramycin · Midecamycin	further macrolide members named in...

Ketolides (Telithromycin) (1)

Telithromycin	respiratory infections due to macrolide-resistant...
----------------------	--

Lincosamides (Lincomycin & Clindamycin) (2)

Clindamycin	anaerobic infections outside the CNS
Lincomycin	gram-positive and anaerobic infections

Streptogramins (Quinupristin / Dalfopristin) (1)

Quinupristin + Dalfopristin ★	VRE
--------------------------------------	-----

Chloramphenicol (1)

Chloramphenicol	empiric meningitis treatment
------------------------	------------------------------

Aminoglycosides (7)

Gentamicin	UTI
Tobramycin	UTI
Amikacin	as tobramycin, plus gram-negatives resistant...
Streptomycin	tuberculosis
Netilmicin · Kanamycin	interchangeable with gentamicin/tobramycin for most...
Neomycin	topical only — minor skin...
Paromomycin	intestinal amebiasis

Tetracyclines (5)

Doxycycline (2nd gen)	CAP
Minocycline (2nd gen) ★	preferred for methicillin-resistant staph
Tetracycline (1st gen)	H. pylori
Demeclocycline ★	SIADH — exploits its own...
Oxytetracycline · Chlortetracycline (1st gen)	eye and skin infections

Glycylcyclines (Tigecycline) (1)

Tigecycline	complicated intra-abdominal infection
--------------------	---------------------------------------

Oxazolidinones (Linezolid) (1)

Linezolid ★	VRE bacteremia
--------------------	----------------

Folate Synthesis Inhibitors (TMP-SMX) (1)

Trimethoprim + Sulfamethoxazole (co-trimoxazole) ★	DOC for Pneumocystis pneumonia —...
---	-------------------------------------

Antimycobacterial Drugs

12 classes · 23 drugs

Isoniazid (INH) (1)

Isoniazid (H, INH)	active TB
---------------------------	-----------

Rifamycins (3)

Rifampin (rifampicin, R)	TB
Rifapentine (P)	long t _{1/2} → weekly latent-TB...
Rifabutin	equally effective as rifampin —...

Pyrazinamide (1)

Pyrazinamide (Z)	active TB — intensive phase...
-------------------------	--------------------------------

Ethambutol (1)

Ethambutol (E)	active TB
-----------------------	-----------

Aminoglycosides *(second-line)* (4)

Streptomycin	MDR-TB
Amikacin	MDR-TB
Kanamycin	MDR-TB
Capreomycin	MDR-TB

Fluoroquinolones *(second-line)* (4)

Moxifloxacin	MDR-TB
Levofloxacin	MDR-TB
Gatifloxacin	MDR-TB
Ofloxacin	MDR-TB

Oral Bacteriostatic Second-Line Agents (3)

Ethionamide	MDR-TB
Cycloserine	MDR-TB
Para-aminosalicylic acid (PAS)	MDR-TB

Bedaquiline (1)

Bedaquiline	XDR-TB and treatment-intolerant MDR-TB —...
--------------------	---

Nitroimidazoles — Pretomanid & Delamanid (2)

Pretomanid	XDR-TB and treatment-intolerant MDR-TB —...
Delamanid	MDR-TB combination regimens, especially where...

Linezolid (1)

Linezolid	XDR-TB / treatment-intolerant MDR-TB —...
------------------	---

Dapsone (1)

Dapsone	leprosy MDT
----------------	-------------

Clofazimine (1)

Clofazimine	leprosy
--------------------	---------

Antifungal Drugs

8 classes · 19 drugs

Polyenes (3)

Amphotericin B	systemic / invasive mycoses —...
Nystatin	non-invasive candidiasis — oropharynx, esophagus,...
Natamycin	fungal eye infection — blepharitis,...

Allylamines (1)

Terbinafine ★	drug of choice for dermatophytoses,...
----------------------	--

Azoles — Imidazoles (2)

Clotrimazole · Miconazole · Econazole · Isoconazole · Sertaconazole · Tioconazole · Luliconazole	topical mycoses — skin, mucosal,...
Ketoconazole	topical mycoses

Azoles — Triazoles (6)

Fluconazole (1st gen)	Candida
Itraconazole (1st gen)	Candida
Voriconazole (2nd gen)	invasive aspergillosis
Posaconazole (2nd gen)	broadest spectrum — the only...
Isavuconazole (2nd gen)	invasive aspergillosis
Oteseconazole (tetrazole)	recurrent vulvovaginal candidiasis

Echinocandins (4)

Caspofungin	invasive candidiasis
Micafungin	invasive candidiasis
Anidulafungin	invasive candidiasis
Rezafungin	invasive candidiasis

Triterpenoids (1)

Ibrexafungerp	vulvovaginal candidiasis
----------------------	--------------------------

Flucytosine (1)

Flucytosine (5-FC) ★	Candida and Cryptococcus infections —...
-----------------------------	--

Griseofulvin

(1)

Griseofulvin ●

dermatophytoses — accumulates in skin,...

Antiparasitic Drugs

18 classes · 36 drugs

Aminoquinolines — Chloroquine, Amodiaquine, Piperaquine

(3)

Chloroquine ★

DOC for P. ovale &...

Amodiaquine

P. falciparum — as artesunate...

Piperaquine

chloroquine-resistant P. falciparum — as...

Quinine & Quinidine

(2)

Quinine

acute blood-stage malaria

Quinidine

acute blood-stage malaria

Mefloquine

(1)

Mefloquine

prevention & treatment of drug-resistant...

Artemisinin & Derivatives (ACTs)

(4)

Artemisinin

uncomplicated falciparum malaria, as an...

Artesunate ★

severe malaria — the workhorse

Artemether

uncomplicated falciparum malaria as an...

Dihydroartemisinin (DHA)

uncomplicated falciparum malaria as an...

Other Heme-Polymerisation Inhibitors

(3)

Lumefantrine

uncomplicated P. falciparum — only...

Pyronaridine

P. falciparum + chloroquine-resistant P...

Halofantrine

falciparum — rapidly effective but...

8-Aminoquinolines — Primaquine & Tafenoquine

(2)

Primaquine ★

DOC for eradicating dormant liver...

Tafenoquine

long-acting analogue

Atovaquone

(1)

Atovaquone + proguanil (Malarone)

chemoprophylaxis

Antimalarial Translation Inhibitors

(2)

Doxycycline / tetracycline ★

chloroquine-resistant P. falciparum with artesunate...

Clindamycin ★

the same role when tetracyclines...

Antimalarial Folate Inhibitors

(2)

Sulfadoxine–pyrimethamine (Fansidar)

blood schizont stages of P...

Proguanil

hepatic pre-erythrocytic forms of P...

Nitroimidazoles — Metronidazole & Tinidazole

(2)

Metronidazole ★

amoebiasis

Tinidazole ★

same spectrum, 2nd-generation — higher...

Pentamidine

(1)

Pentamidine

Pneumocystis pneumonia

Anti-Toxoplasma Agents

(3)

Pyrimethamine ★

toxoplasmosis — combine with sulfadiazine...

Sulfadiazine

toxoplasmosis — combine with pyrimethamine...

Clindamycin

acute toxoplasmosis — the substitute...

Benzimidazoles — Albendazole & Mebendazole

(3)

Albendazole ★

GI nematodes

Mebendazole

GI nematodes only — pinworm,...

Triclabendazole

reserved for liver fluke

Ivermectin

(1)

Ivermectin ●

filarial infections

Neuromuscular Paralysing Anthelmintics

(3)

Pyrantel pamoate

pinworm

Levamisole

common roundworm — less effective...

Piperazine

alternative for roundworm only —...

Praziquantel

(1)

Praziquantel ★

schistosomiasis — the main treatment

Niclosamide

(1)

Niclosamide

tapeworm

Diethylcarbamazine (DEC) (1)

Diethylcarbamazine lymphatic filariasis

Anti-Hepatitis Agents

6 classes · 18 drugs

Interferons (IFN- α / PegIFN) (3)

Interferon alfa-2b	chronic HBV — the only...
Pegylated interferon alfa-2a (PegIFN)	chronic HBV — FDA first-line
Pegylated interferon alfa-2b	chronic HBV / HCV

HBV Nucleoside / Nucleotide Analogues (NAs) (6)

Entecavir (ETV)	chronic HBV — first-line; less...
Tenofovir disoproxil fumarate (TDF)	chronic HBV — first-line; the...
Tenofovir alafenamide (TAF)	chronic HBV — prodrug of...
Lamivudine (LAM)	chronic HBV
Adefovir dipivoxil (ADV)	chronic HBV
Telbivudine (TBV)	chronic HBV

Ribavirin (1)

Ribavirin (RBV) HCV — add-on to PegIFN

HCV NS5B Polymerase Inhibitors (1)

Sofosbuvir (SOF) HCV — all genotypes, always...

HCV NS5A Inhibitors (4)

Velpatasvir (VEL)	HCV genotypes 1–6, fixed-dose with...
Daclatasvir (DCV)	HCV genotype 3 with SOF
Elbasvir (ELB)	HCV genotypes 1a/1b and 4,...
Ledipasvir · Pibrentasvir · Ombitasvir	HCV — components of other...

HCV NS3/4A Protease Inhibitors (3)

Simeprevir (SMV)	HCV genotype 1 or 4,...
Grazoprevir (GRA)	HCV genotypes 1a/1b and 4,...
Voxilaprevir · Glecaprevir · Paritaprevir (ritonavir-boosted)	HCV — components of fixed-dose...

Antivirals & Anti-HIV Drugs

11 classes · 49 drugs

Antitherpes Agents (6)

Acyclovir	HSV-1
Valacyclovir	same — the L-valyl prodrug...
Penciclovir	HSV-1
Famciclovir	same — prodrug of penciclovir,...
Trifluridine	HSV keratitis — ophthalmic, incl...
Docosanol	HSV — topical; mechanism unknown

Anti-CMV Agents (5)

Ganciclovir	CMV retinitis — IV
Valganciclovir	same — the L-valyl prodrug...
Foscarnet	CMV retinitis — incl. nucleoside-analogue-resistant...
Cidofovir	CMV retinitis — long $t_{1/2}$,...
Fomivirsen	CMV retinitis, 2nd line —...

Influenza — M2 Proton Channel Inhibitors (2)

Amantadine	influenza A
Rimantadine	influenza A — fewer neurological...

Influenza — Neuraminidase Inhibitors (3)

Oseltamivir	influenza A & B —...
Zanamivir	influenza A & B —...
Peramivir	influenza A & B —...

Influenza — Polymerase Complex Inhibitors (2)

Baloxavir marboxil ★	influenza A & B —...
Favipiravir	influenza A & B +...

Other Antivirals (3)

Ribavirin	RSV
Palivizumab	RSV prophylaxis in high-risk infants
Imiquimod	HPV

NRTIs — Nucleoside/Nucleotide RT Inhibitors (8)

Abacavir (ABC) ●	HIV — fatal hypersensitivity if...
Zidovudine (AZT / ZDV)	HIV — the first HIV...
Lamivudine (3TC)	HIV — the best-tolerated backbone...
Emtricitabine (FTC)	HIV — skin hyperpigmentation of...
Tenofovir (TDF / TAF)	HIV — renal toxicity, osteomalacia
Didanosine (ddI)	HIV — peripheral neuropathy, pancreatitis
Stavudine (d4T)	HIV — peripheral neuropathy, lipodystrophy
Zalcitabine (ddC)	HIV — peripheral neuropathy, oral...

NNRTIs — Non-Nucleoside RT Inhibitors (5)

Efavirenz (EFV) ★	HIV — CNS effects; bedtime...
Nevirapine (NVP) ●	HIV — serious hepatotoxicity
Rilpivirine (RPV)	HIV — insomnia, depression
Etravirine (ETR)	HIV — after a meal;...
Delavirdine (DLV)	HIV — rash, ↑ liver...

Protease Inhibitors (PIs, -navir) (9)

Atazanavir (ATV)	HIV — indirect hyperbilirubinemia; PR...
Darunavir (DRV)	HIV — a preferred boosted...
Lopinavir/ritonavir (LPV/r)	HIV — co-formulated, boosted
Ritonavir (RTV)	HIV — mostly used as...
Indinavir (IDV) ●	HIV — nephrolithiasis
Saquinavir (SQV)	HIV — QT prolongation, photosensitivity
Fosamprenavir (FPV)	HIV — prodrug of amprenavir
Nelfinavir (NFV)	HIV — GI-heavy, unboosted
Tipranavir (TPV)	HIV — for resistant virus;...

Entry Inhibitors (2)

Maraviroc	HIV — CCR5-tropic only; for...
Enfuvirtide	HIV — a peptide, so...

INSTIs — Integrase Strand Transfer Inhibitors (-gravir) (4)

Dolutegravir (DTG) ★	HIV — first-line in Thailand;...
Bictegravir (BIC)	HIV — co-formulated with FTC...
Raltegravir (RAL)	HIV — twice daily; the...
Elvitegravir (EVG)	HIV — co-formulated with cobicistat...

Antidiabetic Drugs

10 classes · 36 drugs

Insulin (4)

Regular insulin	postprandial
Insulin aspart / lispro / glulisine	postprandial
NPH insulin	basal cover, usually twice daily
Insulin detemir / glargine / degludec	basal cover, once daily

GLP-1 Receptor Agonists (5)

Exenatide · Lixisenatide (exendin-4 derivatives, short-acting)	T2DM — strong postprandial control;...
Liraglutide	T2DM
Dulaglutide	T2DM
Semaglutide (SC or oral)	T2DM
Exenatide-LAR · Albiglutide	T2DM, weekly

Dual GIP/GLP-1 Receptor Agonist (1)

Tirzepatide	T2DM
--------------------	------

DPP-4 Inhibitors (6)

Sitagliptin	T2DM — once daily; 74%...
Vildagliptin	T2DM — the exception: twice...
Saxagliptin	T2DM — short t½ but...
Linagliptin	T2DM — only 3.9% renal...
Alogliptin · Gemigliptin · Teneligliptin · Evogliptin	T2DM — once daily
Trelagliptin	T2DM — once weekly

Sulfonylureas (6)

Glibenclamide (glyburide)	T2DM — 2nd gen; also...
Gliclazide	T2DM — 2nd gen, SUR1-selective
Glipizide	T2DM — 2nd gen
Glimepiride	T2DM — 2nd gen, once...
Gliquidone	T2DM — 2nd gen
Tolbutamide · Chlorpropamide · Tolazamide	1st gen — largely obsolete

Meglitinides (Non-sulfonylureas) (2)

Repaglinide	postprandial hyperglycemia in T2DM —...
Nateglinide	postprandial hyperglycemia in T2DM —...

Biguanide — Metformin (1)

Metformin	first-line type 2 diabetes
------------------	----------------------------

Thiazolidinediones (TZDs) (2)

Pioglitazone	T2DM add-on
Rosiglitazone	T2DM add-on — available in...

α-Glucosidase Inhibitors (3)

Acarbose	postprandial hyperglycemia — barely absorbed
Voglibose	postprandial hyperglycemia — barely absorbed
Miglitol	postprandial hyperglycemia — 50–100% absorbed,...

SGLT2 Inhibitors (6)

Empagliflozin	T2DM
Dapagliflozin	T2DM
Canagliflozin	T2DM
Ertugliflozin	T2DM
Sotagliflozin	T2DM
Bexagliflozin	T2DM — no outcome studies

Estrogens, Progestins & Contraceptives

8 classes · 34 drugs

Estrogens (6)

Estradiol (E2), estrone (E1), estriol (E3)	natural estrogens — replacement therapy,...
Conjugated estrogens	menopausal hormone therapy
Ethinyl estradiol (EE) ★	the contraceptive estrogen — highest...
Estetrol (E4)	newer COC estrogen — much...
Mestranol, quinestrol	synthetic steroidal estrogens
Diethylstilbestrol, chlorotrianisene	non-steroidal synthetic estrogens

Selective Estrogen Receptor Modulators (SERMs) (3)

Tamoxifen	breast cancer — treatment and...
Toremifene	breast cancer
Raloxifene	prevention & treatment of osteoporosis

Estrogen Receptor Antagonists (2)

Clomiphene ★	the "fertility pill" — ovulation...
Fulvestrant	pure estrogen-receptor antagonist

Aromatase Inhibitors (3)

Anastrozole	breast cancer
Letrozole	breast cancer
Exemestane	breast cancer

Progestins (7)

Progesterone	the natural hormone — HRT...
Medroxyprogesterone acetate (MPA)	contraception
Norethindrone (norethisterone)	contraception
Norgestrel / levonorgestrel	contraception
Norgestimate, desogestrel, gestodene	contraception — potent progestin, low...
Cyproterone acetate	anti-androgenic — acne, hirsutism, androgenic...
Drospirenone	anti-androgenic + anti-mineralocorticoid — less...

Antiprogestins (3)

Mifepristone	medical abortion in early pregnancy...
Onapristone	medical abortion — a more...
Ulipristal ★	emergency contraception — single 30...

Progestin-Only Contraceptives (POC)

(7)

Lynestrenol (minipill) ★	daily contraception — late-dose window...
Desogestrel (minipill)	daily contraception — window 12...
Drospirenone (minipill)	daily contraception — window 24...
Medroxyprogesterone acetate (depot IM)	contraception for 3 months
Levonorgestrel (implant)	long-acting contraception, up to 5...
Etonogestrel (implant)	contraception for 3 years
Progestin IUD	local release — contraception for...

Emergency Contraception

(3)

Levonorgestrel ★	1.5 mg once, or 0.75...
Ulipristal	30 mg once — effective...
EE + norgestrel (Yuzpe method)	2 tablets within 72 h,...

Thyroid Hormones & Antithyroid Drugs

6 classes · 21 drugs

Thyroid Hormone Replacement

(4)

Levothyroxine (L-T4) ★	hypothyroidism — drug of choice
Liothyronine (T3)	short-term replacement only — TSH...
Liotrix (T4:T3 = 4:1)	hypothyroidism — not for routine...
Desiccated thyroid (animal gland)	hypothyroidism — largely obsolete

Thionamides

(3)

Methimazole (MMI / thiamazole)	hyperthyroidism, Graves' — more potent
Propylthiouracil (PTU)	hyperthyroidism — preferred in 1st-trimester...
Carbimazole	pro-drug converted to methimazole —...

Anion Inhibitors

(3)

Potassium perchlorate (ClO₄⁻)	iodide-induced hyperthyroidism — the one...
Thiocyanate (SCN⁻)	
Pertechnetate (TcO₄⁻) · Nitrate (NO₃⁻)	

Iodides

(4)

Lugol's solution (5% iodine + 10% KI)	pre-operative preparation
Saturated solution of potassium iodide (SSKI)	pre-operative preparation
Potassium iodide (KI)	endemic goiter prophylaxis
Sodium ipodate / iopanoic acid (iodinated contrast media)	thyroid storm adjunct

Radioactive Iodine (¹³¹I)

(2)

¹³¹I	treatment — the only isotope...
¹²³I	diagnosis only — 24-h uptake...

Adjunctive & Thyroid Storm Therapy

(5)

Propranolol (β-blocker without ISA — agent of choice) ★	tachycardia, tremor, palpitations, AF, hypertension
Atenolol	same symptomatic control
Diltiazem (CCB)	rate control when β-blockers are...
Dexamethasone / hydrocortisone	inhibits T4 → T3
Lithium carbonate	inhibits hormone release — used...

Drugs Regulating Bone-Mineral Homeostasis

10 classes · 28 drugs

Vitamin D Derivatives (5)

Cholecalciferol (D3)	supplementation — more effective at...
Ergocalciferol (D2)	supplementation
Calcidiol / calcifediol (25(OH)D)	supplementation when liver hydroxylation is...
Alfacalcidol (1(OH)D)	CKD — bypasses the failed...
Calcitriol (1,25(OH)₂D)	CKD

Bisphosphonates (5)

Alendronate (PO, ± vitamin D)	osteoporosis — prevention & treatment
Risedronate (PO)	osteoporosis
Ibandronate (PO, IV)	osteoporosis
Zoledronate (IV)	osteoporosis
Pamidronate (IV)	hypercalcemia of malignancy

RANKL Inhibitors (1)

Denosumab (60 mg SC every 6 months)	osteoporosis treatment in women &...
--	--------------------------------------

Estrogens, STEARs & SERMs (5)

Conjugated equine estrogens	prevention of postmenopausal osteoporosis
17β-estradiol · estradiol hemisuccinate	prevention of postmenopausal osteoporosis
Estradiol valerate + norgestrel	as above — the progestogen...
Tibolone (STEAR)	prevention — with menopausal-symptom relief
Raloxifene (SERM)	treatment and prevention — without...

Calcitonin (1)

Synthetic salmon calcitonin (IV, nasal spray)	hypercalcemia
--	---------------

Parathyroid Hormone Analogues (2)

Teriparatide (recombinant PTH1-34, SC)	osteoporosis treatment
PTH1-84 (recombinant human PTH, SC)	hypoparathyroidism

Sclerostin Inhibitors (1)

Romosozumab (210 mg SC monthly × 12 doses)	osteoporosis in postmenopausal women at...
---	--

Vitamin K Analogues & Fluoride (2)

Menatetrenone (PO, with food)	improves bone mass
Sodium fluoride (PO, sucked or chewed)	prophylaxis of dental caries —...

Calcimimetics (2)

Cinacalcet (PO, small molecule, shorter CaSR activation)	2° hyperparathyroidism on dialysis
Etelcalcetide (IV, octapeptide, longer activation via disulfide bond)	2° hyperparathyroidism in dialysis patients...

Phosphate Binders (4)

Calcium carbonate / citrate / acetate	cheap, and supplies calcium —...
Aluminum carbonate	aluminum toxicity
Lanthanum carbonate	partially absorbed
Sevelamer	non-absorbable — the choice in...

Hypothalamic-Pituitary Hormones

9 classes · 25 drugs

GnRH Agonists (4)

Goserelin	prostate cancer
Leuprorelin	prostate cancer
Triptorelin	prostate cancer
Buserelin	prostate cancer

GnRH Antagonists (3)

Degarelix	prostate cancer
Ganirelix	preventing the LH surge during...
Cetrorelix	preventing the LH surge during...

Gonadotropins (6)

Menotropin (hMG)	FSH + LH mixture 1:1,...
Urofollitropin (uFSH)	purified human FSH, from the...
Follitropin α / β (rFSH)	recombinant FSH — follicle stimulation
Corifollitropin α	recombinant FSH, long-acting
Lutropin α (rLH)	recombinant LH — only in...
Choriogonadotropin α (rhCG)	recombinant hCG — triggers ovulation

Dopamine Agonists (3)

Bromocriptine (ergot derivative)	hyperprolactinemia
Pramipexole (non-ergot)	Parkinson's disease — not for...
Rotigotine (non-ergot)	Parkinson's disease — not for...

Oxytocin & Oxytocin Antagonists (2)

Oxytocin (synthetic)	initiation & augmentation of labor
Atosiban (peptide oxytocin-receptor antagonist)	tocolysis in preterm labor

Growth Hormone (1)

Somatropin (recombinant GH)	growth hormone deficiency
------------------------------------	---------------------------

Somatostatin Analogs (3)

Octreotide	acromegaly
Lanreotide	acromegaly
Pasireotide	acromegaly

Vasopressin & Analogs (2)

Vasopressin (synthetic arginine vasopressin, $t_{1/2} \approx 15$ min)	diabetes insipidus
Desmopressin (DDAVP) ($t_{1/2} \approx 2$ h)	diabetes insipidus

Vasopressin Antagonists (1)

Tolvaptan (non-peptide, oral)	SIADH
--------------------------------------	-------

Anabolic-Androgenic Steroids & Anti-Androgens

7 classes · 27 drugs

Testosterone & Testosterone Esters (5)

Testosterone (gel — short duration)	androgen replacement therapy
Testosterone undecanoate (IM, also PO)	androgen replacement — longest interval
Testosterone enanthate (IM)	androgen replacement
Testosterone propionate (IM)	androgen replacement
Testosterone cypionate (IM)	androgen replacement

17 α -Alkylated Testosterone (5)

Methyltestosterone	androgen replacement
Fluoxymesterone	androgen replacement
Oxymetholone	
Stanozolol	
Danazol	

Anabolic Steroids (6)

Mesterolone (PO)	androgen deficiency
Oxymetholone (PO)	aplastic anemia
Stanozolol (IM)	anemia
Stanozolol (PO)	hereditary angioedema
Danazol (PO)	hereditary angioedema
Nandrolone decanoate (IM)	anemia of chemotherapy & chronic...

Steroid Synthesis Inhibitors (3)

Abiraterone (PO)	advanced prostate cancer — always...
Spironolactone (PO)	K ⁺ -sparing diuretic
Ketoconazole (PO)	azole antifungal

5 α -Reductase Inhibitors (3)

Finasteride (selective type II)	BPH
Dutasteride (dual type I + II)	BPH
Dutasteride + Tamsulosin	BPH — shrink the gland

Steroidal Androgen Receptor Antagonists (2)

Cyproterone	reduction of drive in sexual...
Cyproterone + Ethinylestradiol	androgen-dependent disease in women

Non-Steroidal Androgen Receptor Antagonists (3)

Flutamide (1st gen, PO q8h, t_{1/2} 5–6 h)	advanced prostate cancer
Bicalutamide (1st gen, PO daily, t_{1/2} 7 days)	advanced prostate cancer
Enzalutamide (2nd gen, PO daily, t_{1/2} 5 days)	advanced prostate cancer

Adrenal Steroids & Their Antagonists

5 classes · 22 drugs

Mineralocorticoids (1)

Fludrocortisone (PO)	aldosterone replacement — primary adrenal...
-----------------------------	--

Mineralocorticoid Receptor Antagonists (3)

Spirolactone (PO)	hypertension
Finerenone (PO)	chronic kidney disease in type-2...
Drospirenone (PO)	a progestin with MR-antagonist activity...

Glucocorticoids — Systemic (10)

Hydrocortisone (10 mg) ★	drug of choice for cortisol...
Prednisolone (5 mg) ★	drug of choice for non-adrenal...
Triamcinolone (4 mg)	inflammatory / immune disease —...
Dexamethasone (0.5 mg)	inflammatory disease needing a long,...
Hydrocortisone Na⁺ succinate ● ★	IV, IM — drug of...
Dexamethasone sodium	IV, IM — potent systemic...
Methylprednisolone Na⁺ succinate	IV, IM — anti-inflammatory /...
Methylprednisolone acetate	IM, IA, ID — depot;...
Triamcinolone acetonide	IA only — joint inflammation
Betamethasone (dipropionate + disodium phosphate)	IM, IA

Glucocorticoids — Topical & Local (5)

Oral base	Triamcinolone acetonide 0.1%
Rectal	Hydrocortisone acetate + cinchocaine
Optic / otic	Dexamethasone
Intranasal	Budesonide
Inhaled	Budesonide

Glucocorticoid Antagonists (3)

Ketoconazole	Cushing's syndrome — inhibitor of...
Abiraterone	Cushing's syndrome — 17-hydroxylase and...
Pasireotide	Cushing's disease — somatostatin analogue,...

Anticancer Agents I

13 classes · 46 drugs

Alkylating Agents (12)

Mechlorethamine (N-mustard)	Hodgkin's & non-Hodgkin's lymphoma
Cyclophosphamide (N-mustard)	breast
Ifosfamide (N-mustard)	
Chlorambucil (N-mustard)	CLL
Melphalan (N-mustard)	multiple myeloma
Thiotepa (aziridine)	breast
Busulfan (alkyl sulfonate)	CML
Carmustine (BCNU) (nitrosourea)	brain cancer
Lomustine (CCNU) (nitrosourea)	brain cancer
Dacarbazine (triazine)	Hodgkin's
Temozolomide (triazine)	brain cancer
Procarbazine (methylhydrazine)	Hodgkin's & NHL

Platinum Analogs (3)

Cisplatin	broad range of solid tumours...
Carboplatin (2nd gen)	NSCLC, SCLC, breast, bladder, head...
Oxaliplatin (3rd gen)	colorectal cancer

Antifolates (2)

Methotrexate (MTX)	breast
Raltitrexed	polyglutamated → thymidylate synthase inhibition

Ribonucleotide Reductase Inhibitor (1)

Hydroxyurea (HU)	chronic myelogenous leukemia
-------------------------	------------------------------

Pyrimidine Analogs (5)

5-Fluorouracil (5-FU) ★	colorectal
Capecitabine	breast
Cytarabine (Ara-C)	AML
Gemcitabine (dFdC)	pancreatic
Azacitidine	pyrimidine analog

Purine Analogs (4)

6-Mercaptopurine (6-MP)	AML
6-Thioguanine (6-TG)	ALL
Azathioprine (AZA)	prodrug of 6-MP
Fludarabine · Cladribine · Clofarabine · Nelarabine · Pentostatin	purine analogs

Vinca Alkaloids (3)

Vinblastine	Hodgkin's & NHL
Vincristine	ALL
Vinorelbine	NSCLC

Taxanes (3)

Paclitaxel	breast
Docetaxel	breast
Epothilones	same stabilizing mechanism

Topoisomerase I Inhibitors (2)

Topotecan	SCLC
Irinotecan (CPT-11)	colorectal

Topoisomerase II Inhibitors (2)

Etoposide (VP-16-213)	pediatric leukemia
Teniposide (VM-26)	same class, same activity profile

Anthracyclines (5)

Doxorubicin	breast
Daunorubicin	AML
Idarubicin (analog of daunorubicin)	AML
Epirubicin (analog of doxorubicin)	
Mitoxantrone	

Bleomycin & Mitomycin C (3)

Bleomycin	Hodgkin's & NHL
Mitomycin C	superficial bladder
Neocarzinostatin	antitumour antibiotic

L-Asparaginase (1)

L-Asparaginase	standard agent for acute lymphocytic...
-----------------------	---

EGFR Inhibitors — Small-Molecule TKIs (4)

Gefitinib, Erlotinib (1st gen, reversible)	EGFR-mutant NSCLC — L858R, Del19...
Afatinib, Dacomitinib (2nd gen, irreversible)	as above + T790M, wild-type/mutant...
Osimertinib (3rd gen, irreversible)	T790M resistance — L858R, Del19,...
Mobocertinib	exon 20 insertion EGFR

EGFR Inhibitors — Monoclonal Antibodies (3)

Cetuximab (chimeric IgG1)	EGFR-driven solid tumours — binds...
Panitumumab (human IgG2)	EGFR-driven solid tumours — no...
Necitumumab (human IgG1)	targets the S468R variant —...

HER2 Inhibitors (5)

Trastuzumab (mAb IgG1, IV)	HER2+ breast/gastric — blocks ligand-independent...
Pertuzumab (mAb IgG1, IV)	HER2+ breast — blocks ligand-dependent...
Lapatinib (TKI, reversible, oral)	HER1 + HER2
Neratinib (TKI, irreversible, oral)	HER1, HER2, HER4
Tucatinib (TKI, reversible, oral)	HER2, HER3

BCR-ABL Inhibitors (6)

Imatinib (1st gen)	CML — also PDGFR, SCF,...
Nilotinib (2nd gen)	CML — binds the inactive...
Dasatinib (2nd gen)	CML — binds the active...
Bosutinib (2nd gen)	CML — binds both conformations
Ponatinib (3rd gen)	resistant CML — very broad
Asciminib (4th gen)	resistant CML — myristoyl pocket,...

mTOR Inhibitors (3)

Sirolimus (rapamycin)	renal-transplant immunosuppression
Temsirolimus ★	advanced renal cell carcinoma —...
Everolimus	advanced renal cell carcinoma

BRAF Inhibitors (3)

Vemurafenib	BRAF V600E/V600D/V600R mutant tumours
Dabrafenib	BRAF V600E/D/R/K mutant tumours
Encorafenib	BRAF V600E/D/K mutant tumours

PARP Inhibitors (4)

Olaparib	HR-deficient / BRCA-mutated tumours
Rucaparib	HR-deficient / BRCA-mutated tumours
Niraparib	HR-deficient / BRCA-mutated tumours
Talazoparib	HR-deficient / BRCA-mutated tumours

Angiogenesis (VEGF) Inhibitors (8)

Bevacizumab (humanized IgG1 vs VEGF-A)	various cancers — neutralises VEGF-A,...
ziv-Aflibercept (VEGFR1+2 domains fused to IgG1 Fc)	wet AMD
Faricimab (bispecific IgG1: Ang-2 + VEGF-A)	AMD
Ramucirumab (human IgG1 vs VEGFR-2)	solid tumours — t½ ~14...
Sunitinib (multikinase TKI)	VEGFR1-3, PDGFRα/β, KIT, FLT3, RET,...
Axitinib (TKI)	VEGFR1-3 — the selective one
Lenvatinib (TKI)	VEGFR1-3, FGFR1-4, PDGFRα, KIT, RET
Sorafenib (multikinase TKI)	c-CRAF, BRAF, KIT, FLT-3, RET,...

Cancer Vaccines (1)

Sipuleucel-T (autologous, personalized)	prostate cancer — antigen is...
--	---------------------------------

Immune Checkpoint Inhibitors (3)

Ipilimumab (anti-CTLA-4)	releases the priming-phase brake on...
anti-PD-1 antibodies	release the effector-phase brake at...
anti-PD-L1 antibodies	block the tumour's side of...

Anti-Estrogens — SERMs & SERDs (3)

Tamoxifen (SERM)	ER+ breast cancer — the...
Raloxifene (SERM)	ER-mediated disease
Fulvestrant (SERD)	ER+ breast cancer — pure...

Aromatase Inhibitors (3)

Exemestane (Type I, steroidal)	ER+ breast cancer — irreversibly...
Anastrozole (Type II, nonsteroidal)	ER+ breast cancer — reversible...
Letrozole (Type II, nonsteroidal)	ER+ breast cancer — reversible...

Anti-Androgens (3)

Flutamide	advanced prostate cancer
Bicalutamide	advanced prostate cancer
Nilutamide	advanced prostate cancer

Gene Expression Modulators (4)

Vorinostat (HDI)	cutaneous T-cell lymphoma
Romidepsin (HDI)	CTCL, same setting
Retinoids (e.g. tretinoin)	differentiation therapy
Arsenic trioxide	differentiation therapy

Immunomodulators

13 classes · 33 drugs

Induction Antibodies (5)

Rabbit ATG (thymoglobulin)	induction
Alemtuzumab	induction — anti-CD52, depletes T...
OKT3	induction — depletionary
Basiliximab	induction — non-depleting chimeric anti-IL-2Rα
Daclizumab	induction — non-depleting

B-Cell Targeted Therapy (Desensitization) (3)

IVIG (low dose + plasmapheresis)	desensitization — high-antibody recipient, ABO-incompatible...
Rituximab	B-cell depletion
Bortezomib	plasma-cell directed

Corticosteroids (2)

Dexamethasone	induction phase, IV, first post-op...
Prednisone	maintenance — oral, tapered down...

Calcineurin Inhibitors (2)

Cyclosporine	maintenance immunosuppression after transplant
Tacrolimus	same — the more potent...

Antimetabolites (2)

Azathioprine	maintenance immunosuppression — no longer...
Mycophenolic acid / mycophenolate	maintenance, as the adjunct to...

mTOR Inhibitors (2)

Sirolimus	maintenance immunosuppression
Everolimus	maintenance immunosuppression

Costimulation Blocker (1)

Belatacept	maintenance immunosuppression in renal transplant...
-------------------	--

JAK Inhibitors (1)

Tofacitinib	rheumatoid arthritis
--------------------	----------------------

Specific Immunoglobulins (6)

Rabies immunoglobulin (ERIG equine / HRIG human)	post-exposure prophylaxis — transdermal bites...
Rho(D) immunoglobulin ★	prevention of Rh hemolytic disease...
Tetanus immunoglobulin	tetanus
CMV immunoglobulin	cytomegalovirus
Anti-snake venom immunoglobulin	envenomation
Anti-thymocyte immunoglobulin	transplant induction and rejection

Myeloid Growth Factors & Erythropoietins (4)

Filgrastim (rhG-CSF)	neutropenia from myelosuppressive chemotherapy, bone...
rhGM-CSF	neutropenia after BMT and antiviral...
Epoetin alfa	anemia of renal failure or...
Darbepoetin alfa	same — long-acting, ~3× epoetin

Interferons & Imiquimod (1)

Imiquimod (topical)	wart
----------------------------	------

Immunomodulatory Imides (2)

Thalidomide ●	withdrawn — birth defects
Lenalidomide	multiple myeloma

CAR T-Cell Therapy (2)

Tisagenlecleucel	CD19-directed, lentiviral, 2nd gen —...
Axicabtagene ciloleucel	CD19-directed, retroviral, 2nd gen —...

Pharmacology 3

ภท 401 · ภาคต้น · CNS & neuropsychiatry — psychiatry → neurology

Drugs Acting on GABA-A Receptor

4 classes · 18 drugs

Benzodiazepines (10)

Triazolam	early insomnia
Midazolam	early insomnia
Temazepam	late insomnia
Lorazepam	late insomnia
Diazepam	status epilepticus
Clonazepam	panic disorder
Alprazolam	panic disorder
Chlordiazepoxide	anxiety
Clorazepate	anxiety
Clobazam	febrile seizure

Barbiturates (3)

Phenobarbital	CNS depressant / anticonvulsant
Pentobarbital	CNS depressant
Thiopental	CNS depressant / anaesthesia

Z-Drugs (4)

Zolpidem	early insomnia
Zaleplon	insomnia
Indiplon	insomnia
Zopiclone	insomnia

Flumazenil (1)

Flumazenil	benzodiazepine overdose
-------------------	-------------------------

Analgesics I — Opioids & Paracetamol

5 classes · 14 drugs

Weak Opioids (2)

Codeine	mild–moderate pain
Tramadol	mild–moderate pain

Strong Opioids (6)

Morphine	moderate–severe pain — the reference...
Oxycodone	moderate–severe pain
Methadone	moderate–severe pain — very long...
Meperidine (pethidine)	acute pain only — not...
Fentanyl (and sufentanil, remifentanyl)	severe pain — very potent...
Heroin	

Partial Agonists & Mixed Agonist–Antagonists (3)

Buprenorphine	pain — partial μ agonist,...
Nalbuphine	pain — μ antagonist, strong...
Pentazocine	pain — partial μ agonist...

Opioid Antagonists (2)

Naloxone	opioid overdose
Naltrexone	alcoholism

Paracetamol Overdose & Antidote (1)

N-acetylcysteine	paracetamol overdose — replenishes the...
-------------------------	---

CNS Stimulants and Drugs Affecting Appetite

slides not released

 **Nothing here yet — the lecturer has not released these slides.**

อ.วิลาสินี (รศ.ดร.วิลาสินี หิรัญพานิชย์ ขาโตะ) · week 3 — Wednesday 19 ส.ค. 69, 10:00–12:00, R207
Examined material; the gap is real, not an omission.

Drug Abuse and Addiction

slides not released

 **Nothing here yet — the lecturer has not released these slides.**

อ.อนันต์ชัย (อ.ดร.อนันต์ชัย อัศวเมธี) · week 4 — Wednesday 26 ส.ค. 69, 10:00–12:00, R207
Examined material; the gap is real, not an omission.

Anxiolytics and Hypnotics

7 classes · 30 drugs

Antidepressants for Anxiety (12)

Fluoxetine (SSRI)	PD
Sertraline (SSRI)	PD, SAD
Citalopram (SSRI)	off-label only: GAD, PD, SAD
Escitalopram (SSRI)	GAD
Paroxetine (SSRI)	PD, SAD, GAD — the...
Fluvoxamine (SSRI)	off-label only: GAD, PD, SAD
Duloxetine (SNRI)	GAD
Venlafaxine XR (SNRI)	GAD
Desvenlafaxine (SNRI)	off-label only: GAD, PD, SAD
Clomipramine · Imipramine · Desipramine · Nortriptyline (TCA)	off-label only: GAD, PD, SAD
Phenelzine (MAOI)	off-label only: GAD, PD, SAD
Mirtazapine (5-HT₂, 5-HT₃, α₂, H₁ antagonist)	off-label only: anxiety, GAD, PD,...

Benzodiazepines in Anxiety (5)

Clonazepam	PD
Alprazolam	anxiety, PD
Lorazepam	anxiety
Chlordiazepoxide	anxiety
Oxazepam	anxiety

Pregabalin & Gabapentin (2)

Pregabalin	off-label: GAD, SAD
Gabapentin	off-label: GAD, SAD, PD

Buspirone (1)

Buspirone	anxiety
------------------	---------

Other Agents for Anxiety (5)

Propranolol (β₁, β₂ antagonist)	off-label: anxiety, PD, SAD —...
Hydroxyzine (H₁ antagonist)	anxiety
Trifluoperazine (D₂ antagonist)	anxiety
Olanzapine · Quetiapine (D₂, 5-HT₂, H₁ antagonist)	off-label: anxiety, GAD
Etifoxine	anxiety — neurosteroid-site APL, see...

Melatonin & Melatonin Receptor Agonists (2)

Melatonin (prolonged release)	primary insomnia in the elderly
Ramelteon	insomnia — MT ₁ /MT ₂ agonist

Orexin Receptor Antagonists (DORAs) (3)

Lemborexant	insomnia
Suvorexant	insomnia
Daridorexant	insomnia

Antidepressants and Lithium

8 classes · 25 drugs

MAO Inhibitors (MAOIs and RIMAs) (3)

Tranylcypromine (irreversible)	depression
Iproniazid (irreversible)	depression
Moclobemide (RIMA)	depression — MAO-A selective at...

Tricyclic Antidepressants (TCAs) (7)

Amitriptyline	depression
Nortriptyline	depression
Clomipramine	depression
Imipramine	depression
Maprotiline	depression — notable seizure risk
Doxepin	depression
Melitracen (with flupentixol)	depression

SSRIs (5)

Fluoxetine ★	depression — the long-half-life one
Fluvoxamine	depression
Paroxetine	depression — anticholinergic at high...
Sertraline	depression
Escitalopram (and citalopram)	depression

SARIs — Serotonin Antagonist and Reuptake Inhibitors (1)

Trazodone	depression
------------------	------------

SNRIs (5)

Venlafaxine	depression
Duloxetine	depression
Desvenlafaxine	depression
Milnacipran	depression
Levomilnacipran	depression

NDRIs (1)

Bupropion	depression
------------------	------------

NaSSAs (2)

Mirtazapine	depression
Mianserin	depression

Lithium (1)

Lithium ★	acute manic episode
------------------	---------------------

Antipsychotics

3 classes · 25 drugs

Conventional (First-Generation) Antipsychotics (9)

Chlorpromazine	psychotic disorders
Thioridazine	psychotic disorders — most likely...
Perphenazine	psychotic disorders
Zuclopenthixol	psychotic disorders
Haloperidol ●	psychotic disorders
Flupenthixol	psychotic disorders
Fluphenazine	psychotic disorders
Pimozide	psychotic disorders
Trifluoperazine	psychotic disorders

Extrapyramidal Side Effects (3)

Benztropine	acute EPS — dystonia, parkinsonism
Trihexyphenidyl	acute EPS — dystonia, parkinsonism
Diphenhydramine	acute EPS — dystonia, parkinsonism

Atypical (Second-Generation) Antipsychotics (13)

Risperidone (SDA)	psychotic disorders — high prolactin...
Paliperidone (SDA)	psychotic disorders — the active...
Clozapine (SDA) ●	psychotic disorders — agranulocytosis, myocarditis,...
Olanzapine (SDA)	psychotic disorders
Quetiapine (SDA)	psychotic disorders
Ziprasidone (SDA)	psychotic disorders — QTc
Lurasidone (SDA)	psychotic disorders
Aripiprazole (D2 partial agonist)	psychotic disorders
Brexpiprazole (D2 partial agonist)	psychotic disorders
Cariprazine (D2 partial agonist)	psychotic disorders
Amisulpride (combined D2/D3 antagonist)	psychotic disorders — QTc
Lumateperone (DA, 5-HT and glutamate modulator)	psychotic disorders
Pimavanserin (5-HT2A antagonist) ★	psychosis in Parkinson's disease —...

Local and General Anesthetics

slides not released

⌚ Nothing here yet — the lecturer has not released these slides.

อ.อาณัติชัย (ผศ.ดร.อาณัติชัย มายูเทศ) · week 9 — Wednesday 14 ต.ค. 69, 10:00–12:00, R207

Examined material; the gap is real, not an omission.

Neurodegenerative Disease Drugs

8 classes · 26 drugs

Cholinesterase Inhibitors (ChEIs) (4)

Donepezil	AD — mild, moderate and...
Rivastigmine	AD — mild, moderate and...
Galantamine	AD — mild to moderate
Ginkgo biloba standardized extract (EGb761)	AD

NMDA Receptor Antagonist — Memantine (1)

Memantine ★	moderate to severe AD —...
--------------------	----------------------------

Anti-Amyloid Monoclonal Antibodies (3)

Lecanemab	early AD — mild dementia...
Aducanumab	AD — anti-amyloid mAb
Donanemab	AD — anti-amyloid mAb

Levodopa + Dopa Decarboxylase Inhibitor (2)

Levodopa + carbidopa	PD — the standard combination
Levodopa + benserazide	PD — the standard combination

MAO-B Inhibitors (3)

Selegiline	early PD
Rasagiline	early PD
Safinamide	PD — the reversible one

COMT Inhibitors (1)

Entacapone	PD — add-on to levodopa...
-------------------	----------------------------

Dopamine Receptor Agonists (9)

Bromocriptine (ergot)	PD
Cabergoline (ergot)	PD
Pergolide (ergot)	PD
Lisuride (ergot)	PD
Pramipexole (non-ergot) ★	PD — also depression in...
Ropinirole (non-ergot)	PD
Rotigotine (non-ergot)	PD
Piribedil (non-ergot)	PD
Apomorphine (non-ergot)	PD

Amantadine & Anticholinergics (3)

Amantadine	PD symptoms
Trihexyphenidyl	PD — resting tremor, rigidity...
Benzotropine	PD — resting tremor, rigidity...

Antiepileptics

7 classes · 26 drugs

Na⁺ Channel Blockers (9)

Phenytoin, fosphenytoin	epilepsy — focal & generalized...
Carbamazepine	epilepsy — focal & generalized...
Oxcarbazepine	epilepsy
Eslicarbazepine acetate	epilepsy
Lamotrigine	epilepsy
Lacosamide ★	epilepsy — the slow-inactivation agent
Rufinamide	epilepsy
Zonisamide	epilepsy
Topiramate, valproate	also block VGSC — see...

Ca²⁺ Channel Blockers (5)

Gabapentin	epilepsy
Pregabalin ★	epilepsy
Mirogabalin	neuropathic pain only — not...
Ethosuximide	absence seizure
Valproate	broad spectrum, incl. absence

GABA Enhancers (7)

Phenobarbital	epilepsy — focal, generalized tonic-clonic
Clonazepam	epilepsy — incl. absence, atonic/tonic,...
Clobazam	epilepsy — focal onset
Diazepam, midazolam	status epilepticus
Nitrazepam	atonic / tonic seizure
Vigabatrin	epilepsy — GABA-T inhibitor
Tiagabine	epilepsy — GAT-1 inhibitor

Broad Spectrum — Valproate (1)

Sodium valproate / valproic acid	epilepsy — broad spectrum, incl....
---	-------------------------------------

Broad Spectrum — Topiramate (1)

Topiramate ★	epilepsy — broad spectrum
---------------------	---------------------------

AMPA Receptor Blocker — Perampanel (1)

Perampanel	epilepsy — AMPA receptor blocker
-------------------	----------------------------------

SV2A Modulators — Levetiracetam (2)

Levetiracetam	epilepsy — focal, generalized tonic-clonic,...
Brivaracetam	epilepsy — SV2A modulator

Migraine Headache

4 classes · 15 drugs

Ergots (2)

Ergotamine (usually with caffeine)	acute migraine attack — abortive
Dihydroergotamine (DHE)	acute migraine attack — abortive

Triptans & Lasmiditan (5)

Sumatriptan	acute migraine — abortive
Eletriptan	acute migraine — CYP3A4, so...
Almotriptan · frovatriptan · naratriptan · rizatriptan · zolmitriptan	acute migraine — abortive
Sumatriptan 50 mg + naproxen 250–500 mg	acute migraine — the combination...
Lasmiditan	acute migraine — selective 5-HT1F...

Gepants — CGRP Receptor Antagonists (4)

Rimegepant ★	both — acute migraine with...
Ubrogepant	acute migraine with or without...
Zavegepant	acute migraine with or without...
Atogepant	prevention of episodic and chronic...

Anti-CGRP Monoclonal Antibodies (4)

Erenumab (human IgG2, anti-receptor)	prevention of episodic / chronic...
Fremanezumab (humanised IgG2, anti-CGRP)	prevention of episodic / chronic...
Galcanezumab (humanised IgG4, anti-CGRP) ★	prevention of episodic / chronic...
Eptinezumab (humanised IgG1, anti-CGRP)	prevention of episodic / chronic...

Analgesics II — Two Separate Territories

11 classes · 25 drugs

Sodium Channel Blockers (3)

Carbamazepine	neuropathic pain — especially paroxysmal...
Oxcarbazepine	neuropathic pain — same target
Lidocaine (transdermal patch)	localised neuropathic pain — applied...

Gabapentinoids (α₂δ-1 Ligands) (3)

Gabapentin	neuropathic pain
Pregabalin	neuropathic pain
Mirogabalin	neuropathic pain

Antidepressants — TCAs & SNRIs (5)

Amitriptyline (TCA)	neuropathic pain — NET/5-HTT +...
Nortriptyline (TCA)	neuropathic pain — same triple...
Duloxetine (SNRI)	neuropathic pain
Venlafaxine (SNRI)	neuropathic pain
Milnacipran (SNRI)	neuropathic pain

Opioids — Tramadol & Methadone (2)

Tramadol	neuropathic pain — μ agonist...
Methadone	neuropathic pain — μ agonist...

Other Agents — Capsaicin, Ketamine, Cannabinoids (3)

Capsaicin (topical, ≥ 0.075%)	localised neuropathic pain — TRPV1...
Ketamine	neuropathic pain — NMDA antagonist
Cannabinoids	neuropathic pain

Orphenadrine (2)

Orphenadrine	muscle spasm — 100 mg...
Orphenadrine + paracetamol	muscle spasm with pain —...

Tolperisone & Eperisone (2)

Tolperisone	muscle spasm — t½ 1.5...
Eperisone	muscle spasm — t½ 1.5–2...

Benzodiazepines — Diazepam & Clonazepam (2)

Diazepam	muscle spasm / spasticity —...
Clonazepam	muscle spasm / spasticity —...

Baclofen (1)

Baclofen	spasticity — start 5 mg...
-----------------	----------------------------

Tizanidine (1)

Tizanidine	spasticity / muscle spasm —...
-------------------	--------------------------------

Riluzole ★

approved for spasm in
amyotrophic...